Project Report

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Assignment 2 for COMP7023 Predictive Analytics
School of Computer, Data and Mathematical Sciences
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1 Introduction

Global warming has been the hot topic in the World Health Organisation (WHO) for many years. Vehicle emissions are a major source of environmental pollution and climate change, with carbon dioxide (CO2) and nitrogen oxides (NOX) being among the most significant pollutants. According to European Environment Agency (2023), CO2 has impacted the increase of temperature, while NOX emissions are associated with smog formation, acid rain, and adverse respiratory effects. Hence, the Intergovernmental Panel on Climate Change (IPCC) has suggested to closely monitor the gas emissions from the road vehicles.

This report analyses emissions using the data from the UK Vehicle Certification Agency (VCA), which provides verified measurements of vehicle fuel consumption and emissions under the Worldwide Harmonised Light Vehicle Test Procedure (WLTP). The VCA dataset ensures consistency and accuracy across vehicle testing in accordance with European and international standards (UK Vehicle Certification Agency 2024). Through statistical and predictive modelling, this study explores the key factors influencing emission levels to support sustainable transportation research and evidence-based environmental policymaking.

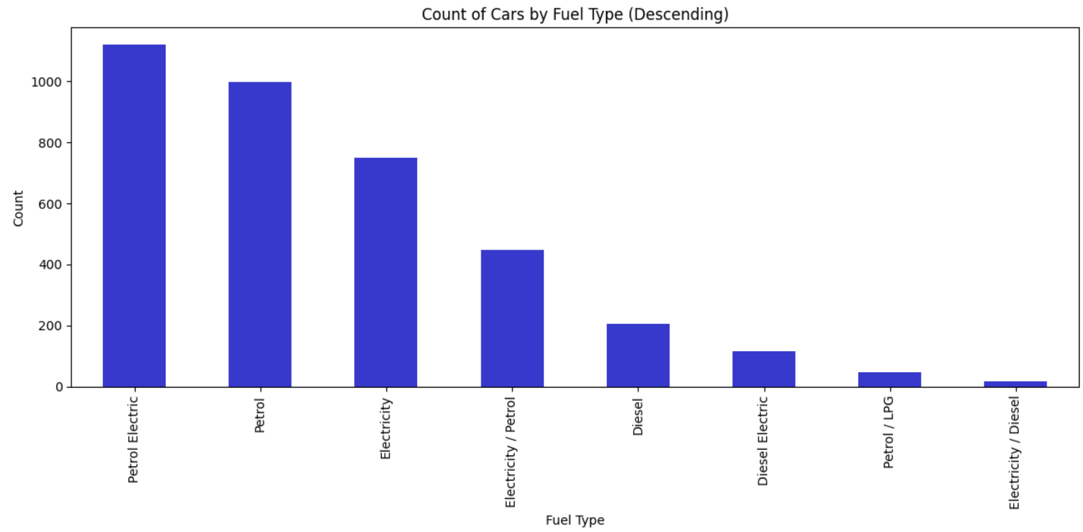
2 Exploratory Data Analysis

The data set contains of 3702 vehicles from different companies and 44 columns of measurement of vehicle types, fuel consumption and emmission.

Category	Columns
Vehicle	Manufacturer, Model, Description, Transmission, Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS), Engine Power (Kw)
Fuel Consumption	Diesel VED Supplement, Testing Scheme, WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined, WLTP Imperial Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted)
Electric Consumption	Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard	Euro Standard
Emmission	WLTP CO2, WLTP CO2 Weighted, Emissions CO [mg/km], THC Emissions [mg/km], Emissions NOx [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date	Date of change

Category	Columns

Out of the 44 columns only 9 columns which are Manufacturer, Model, Description, Transmission, Manual or Automatic, Fuel Type, Powertrain, Euro Standard, and Testing Scheme are the non-numeric data type, the rest of columns are numeric or boolean which can be used modelling. From following chart, we can see the distribution of fuel type of the vehicle in the data set. We can group the vehicles into three main categories: Fuel, Hybrid and Electric.

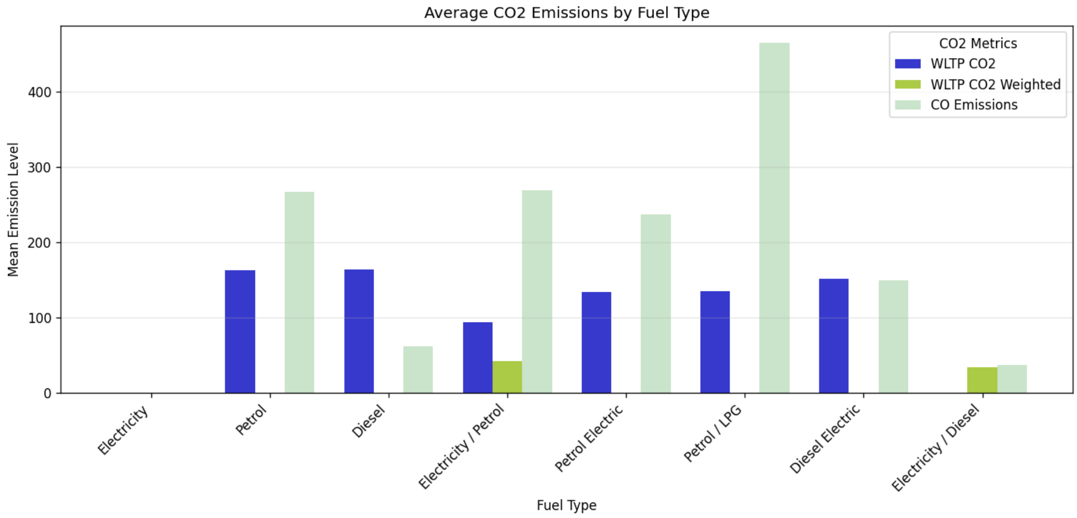


2.1 Emissions Analysis

As the discussion is about emission, we will further understand the different types of emissions such as Carbon Monoxide (CO), Carbon Dioxide (CO2), Nitrogen Oxides (NOx) and Total Hydrocarbons (THC). These emissions are not polluting the environment but impacting our health. According to WHO (2021), when CO binds with haemoglobin more readily than oxyegen, it might reduce oxygen transport in blood and posing serious health risks to humans. While CO2 is non-toxis to humans but it significantly contributes to global warming and climate change as it can trap the heat in the atmosphere (Intergovernmental Panel on Climate Change [IPCC], 2021). When the temperature increase, it might bring disasters to human such as flood or extreme heat wave.

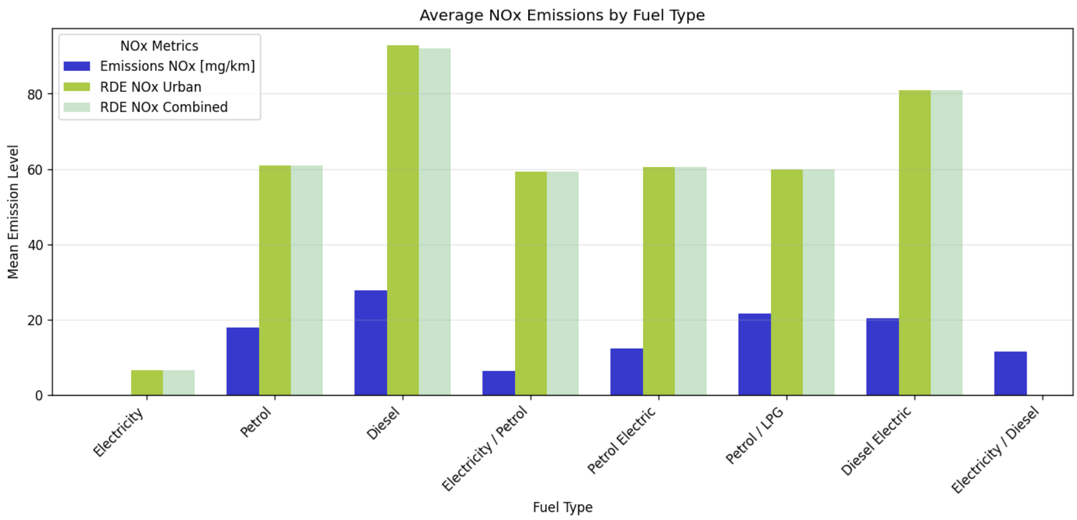
2.1.1 Carbon Monoxide (CO) and Carbon Dioxide (CO2)

From the following bar chart, we can clearly see the emissions of CO2 and CO are co-exists in petrol or hybrid vehicles but not in electric car. Emission of CO is high especially in diesel vehicle and emission of CO2 based on WLTP CO2 is about the same in petrol or diesel verhicles but low in hybrid vehicles. WLTP CO2 Weighted seems to be not reliable as it is empty for most of the vehicles type.



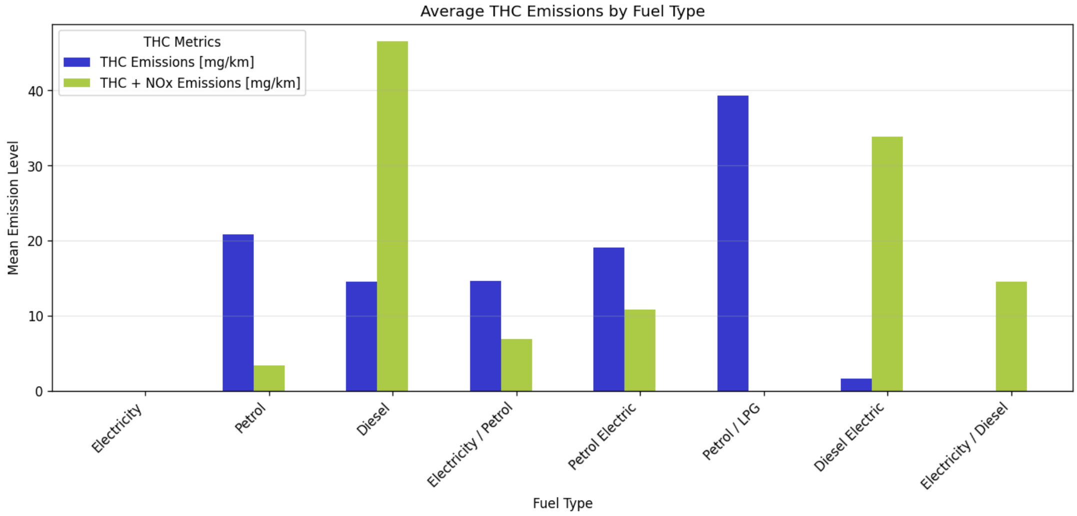
2.1.2 Nitrogen Oxides (NOx)

Nitrogen Oxides (NOx) emissions are a major cause to photochemical smog, acid rain, and respiratory health issues (U.S. Environmental Protection Agency [EPA], 2023). Based on following chart, we find NOx emission exists in all the vehicles including electic car. Electric cars do produce a small average amount of NOx according Veza et al (2023). However, it should be understood that the NOx production from electric vehicles is caused from the source of electric used to charge the car batteries. NO emissions from electric vehicles originate from the fossil fuel–based electricity used to charge them. The mean for NOx in Real Driving Emissions (RDE) for urban and combined are about same but they are higher than Emissions (NOx) [mg/km] which was the result tested in lab.



2.1.3 Total Hydrocarbons (THC)

Total Hydrocarbons (THC) are volatile organic compounds (VOCs) that react with NOx in the presence of sunlight to form ground-level ozone and smog (European Environment Agency [EEA], 2022). THC might cause photochemical pollution and impact the urban air quality but it have to react with NOx, hence, in this report we will not focus in analysing the relationship of THC with other features.



2.2 Target Variables

From the analysis, we have enough data for modelling accross different fuel types but we will choose CO2 as global warming is most common topic when discussing about the impacts of emission. As mentioned above, WLTP CO2 Weighted is lacking of data distribution accross fuel types, hence it will be excluded from target variables. Besides of CO2, we will pick Emissions NOx [mg/km] as our target variables because it has difference mean across fuld type which is easier for our comparison and prediction afterward in this report.

Based on the graphs above, eletric cars have no CO2 emissions, and low NOx emissions. Therefore, all of the eletric cars will be excluded from the datasets when we build the model. Given the distribution, two datasets will be created based on their fuel types - fuel cars only (data_fuel), and hybrid cars only (data_hybrid). Furthermore, since the target variables (WLTP CO2 and NOx emissions) are not highly included in Eletricity /Diesel Fuel type, they will be removed from the data_hybrid.

3 Data Preprocessing

3.1 Model Building

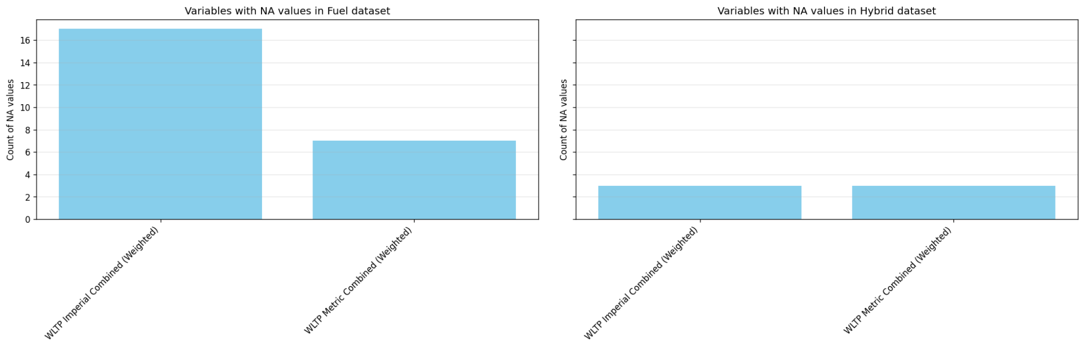
After identified the target variables - Emissions NOx [mg/km] and WLTP CO2, based on the category that we have defined above, we will choose what are the relevant columns to keep in the modelling and remove all irrelevant columns. Reducing the variables/columns can help in improving the performance and cost effectiveness. Variables that are related to the manufacturer or car model will be removed as these variables cannot be used in predictions. As mentioned above, we will create 2 datasets - data_fuel and data_hybrid, which contains fuel vehicles and hybrid vehicles. Hence, the variables that are not relevant to fuel or hybrid will be removed too. For emmissions, we will keep only Emissions NOx [mg/km] and WLTP CO2 as our target variables and drop rest of the emissions related columns. Lastly, date and standard category columns will be removed too as this information cannot be used for modeling.

Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined, WLTP Imperial Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted)	Diesel VED Supplement, Testing Scheme
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard		Euro Standard
Emmission	Emissions NOx [mg/km], WLTP CO2	WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change

After removing the irrelevant variables, we have 19 variables remained in the datasets. Next, we will split our data into two groups as mentioned above which are the hybrid vehicles and fuel vehicles. We are getting **1686 hybrid** vehicles and **1250 fuel** vehicles in our datasets after splitting by the fuel type.

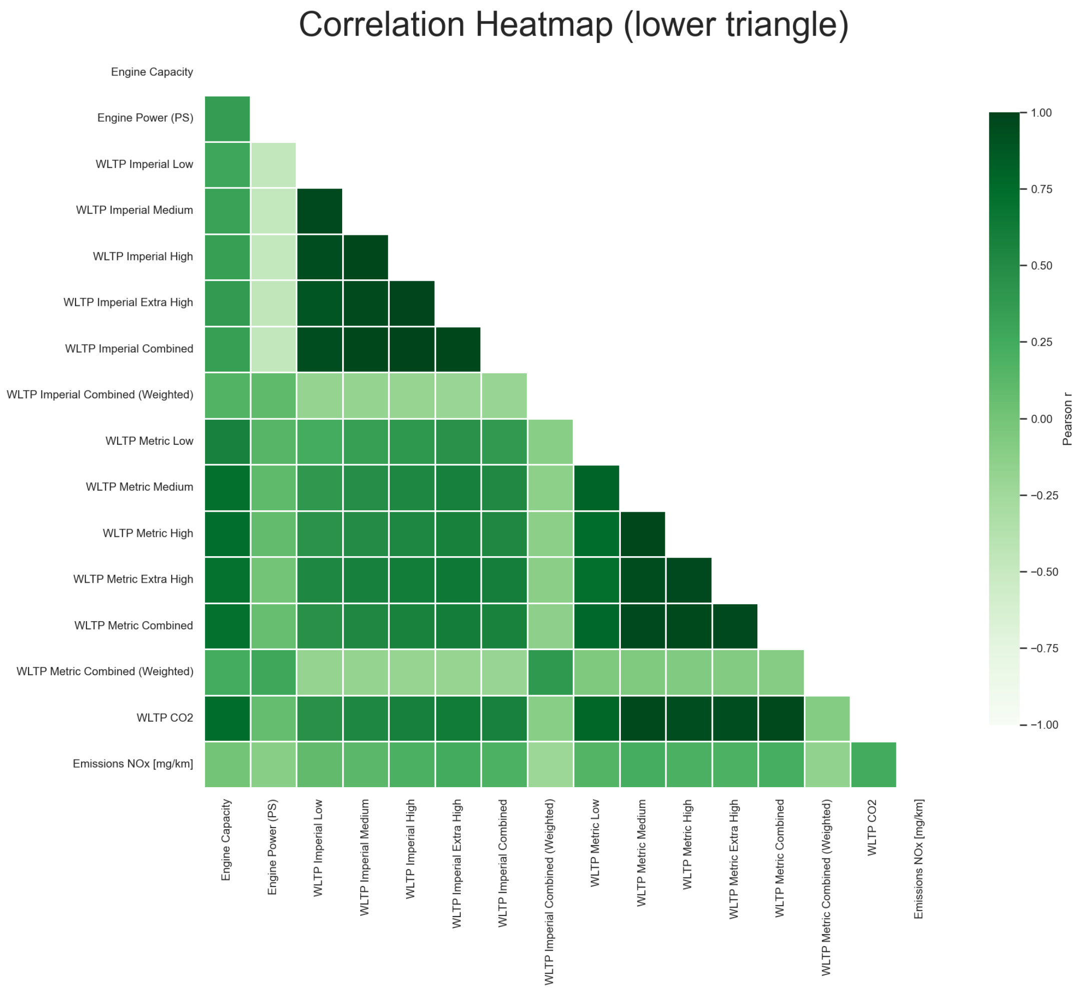
3.2 Data Cleaning

Before we start the modelling, cross checking on the data with "not available (NA)" values or empty, this will help to ensure the accuracy of our model. Based on the result, we found NA values in WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) columns in both fuel and hybrid datasets. After removing the NA rows, we have **1233 fuel** vehicles and **1683 hybrid** vehicles remained in the datasets.

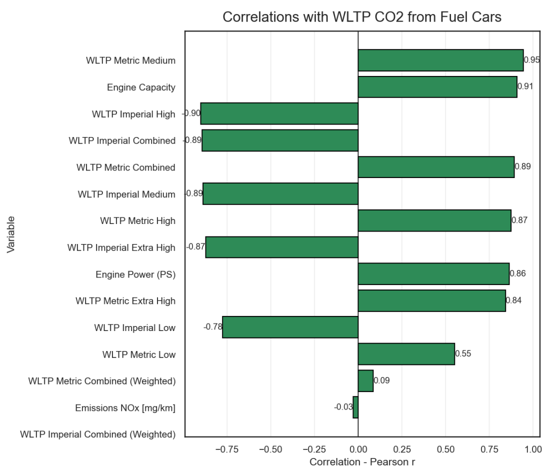


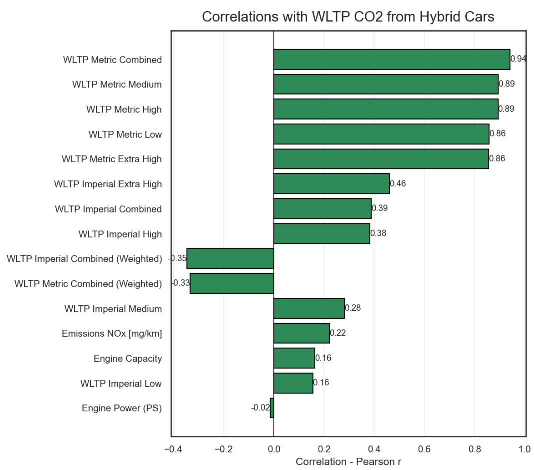
3.3 Correlation Analysis

Correlation analysis started with a heatmap to understand the correlation between each variables. From the heatmap, we can see the correlation with our target variables are low at Engine Power (Ps), WLTP Imperial Combined (Weighted), and WLTP Metric Combined (Weighted).

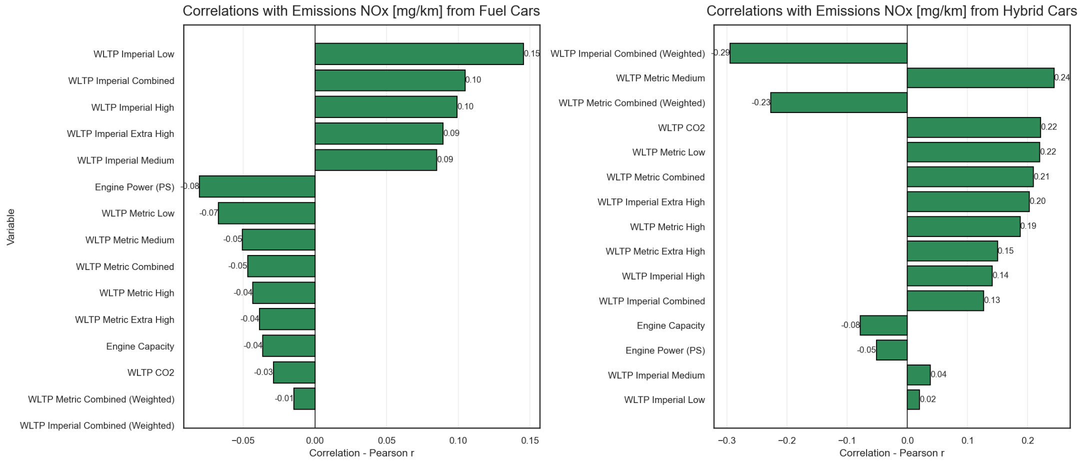


When drilling down to the correlations with CO2, we can see that the WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are having the lowest score in both fuel and hybrid vehicles. Whereas fuel is having high correlation with fuel vehicle but not the hybrid vehicle.



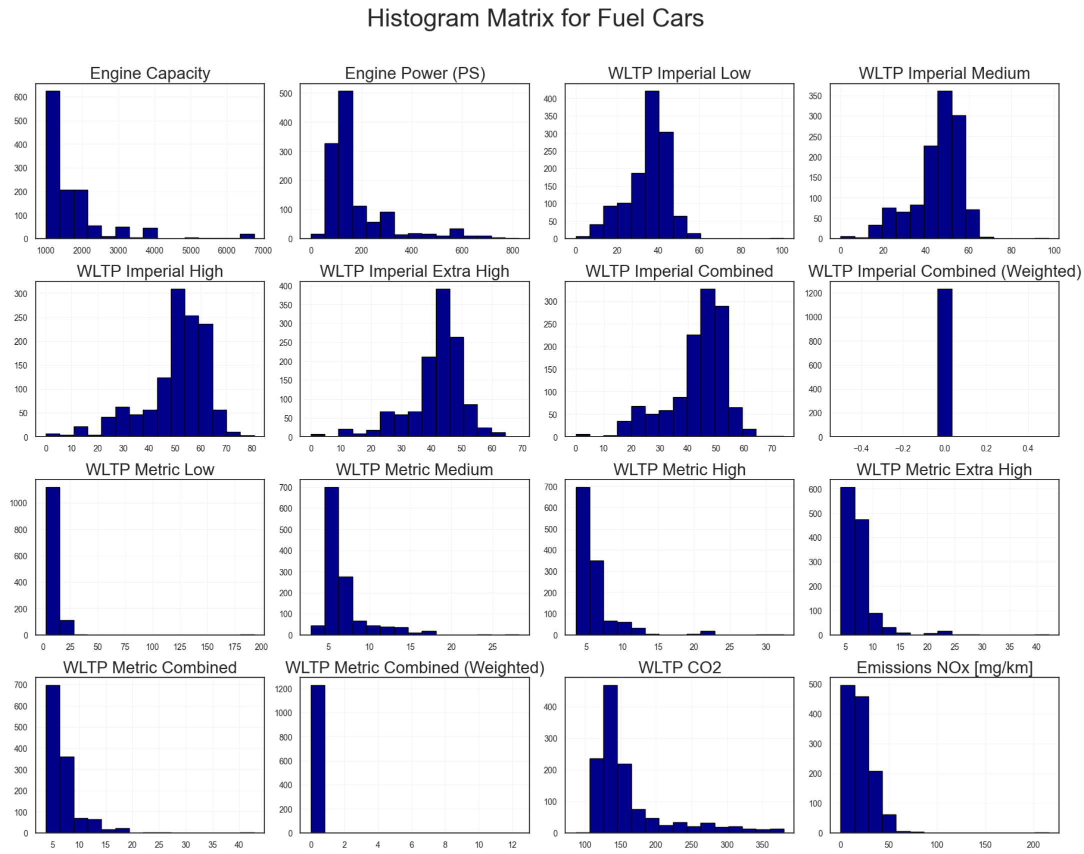


In NOx emmissions, WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are still having lowest correlation in fuel vehicles but these variables are the top three correlated variables in hybrid vehicles. This could be due to the hybrid engine turns on intermittently, it often operates under non-steady, high-load conditions to recharge the battery or accelerate.

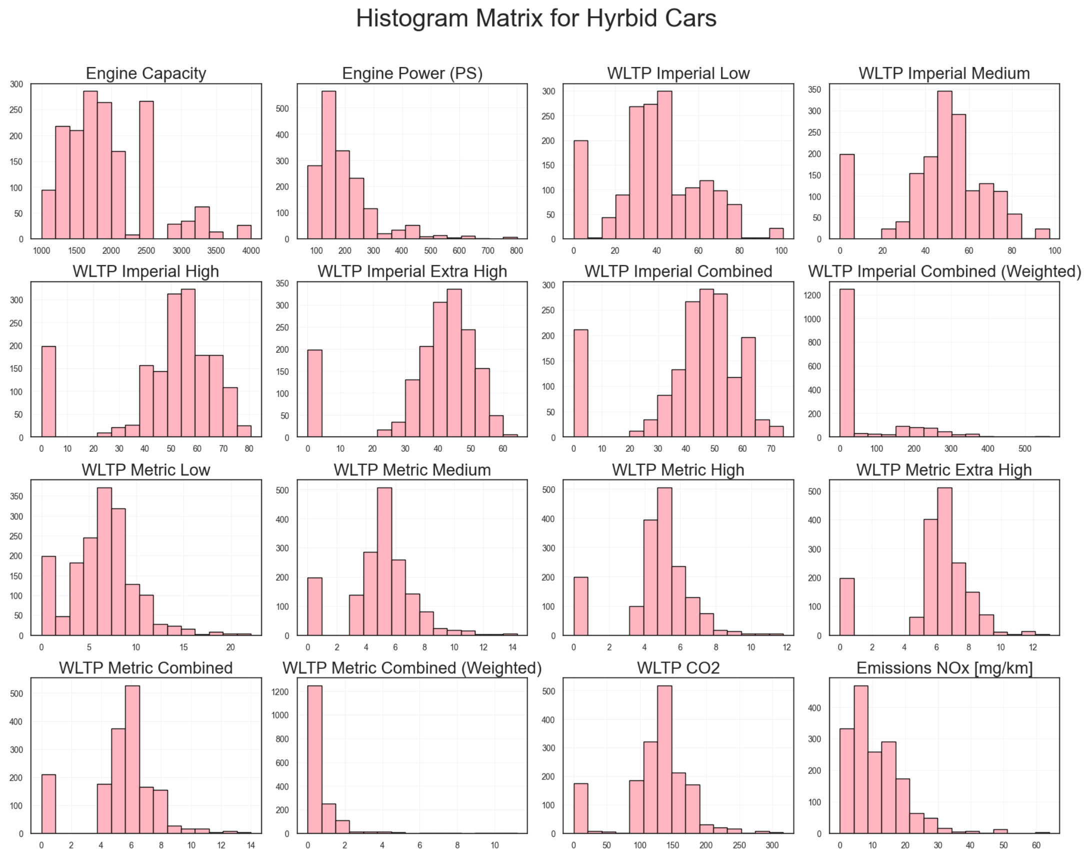


3.4 Data Distribution Analysis

Distribution of data is important in identifying the normalisation or standardisation of each variables in our datasets. Below histogram matrix that showing the data distribution for fuel vehicles, we can see that WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) are not statistically normalised, as their disbutions are tightly clustered and do not follow a standardisation or z-score normalisation. Hence, we can safely remove WLTP Imperial Combined (Weighted) and WLTP Metric Combined (Weighted) columns from *data_fuel* dataset



Each variable in hybrid vehicles is statistically normalised, we do not see abnormal data distribution here. Hence, no changes to the variables for hybrid vehicles.

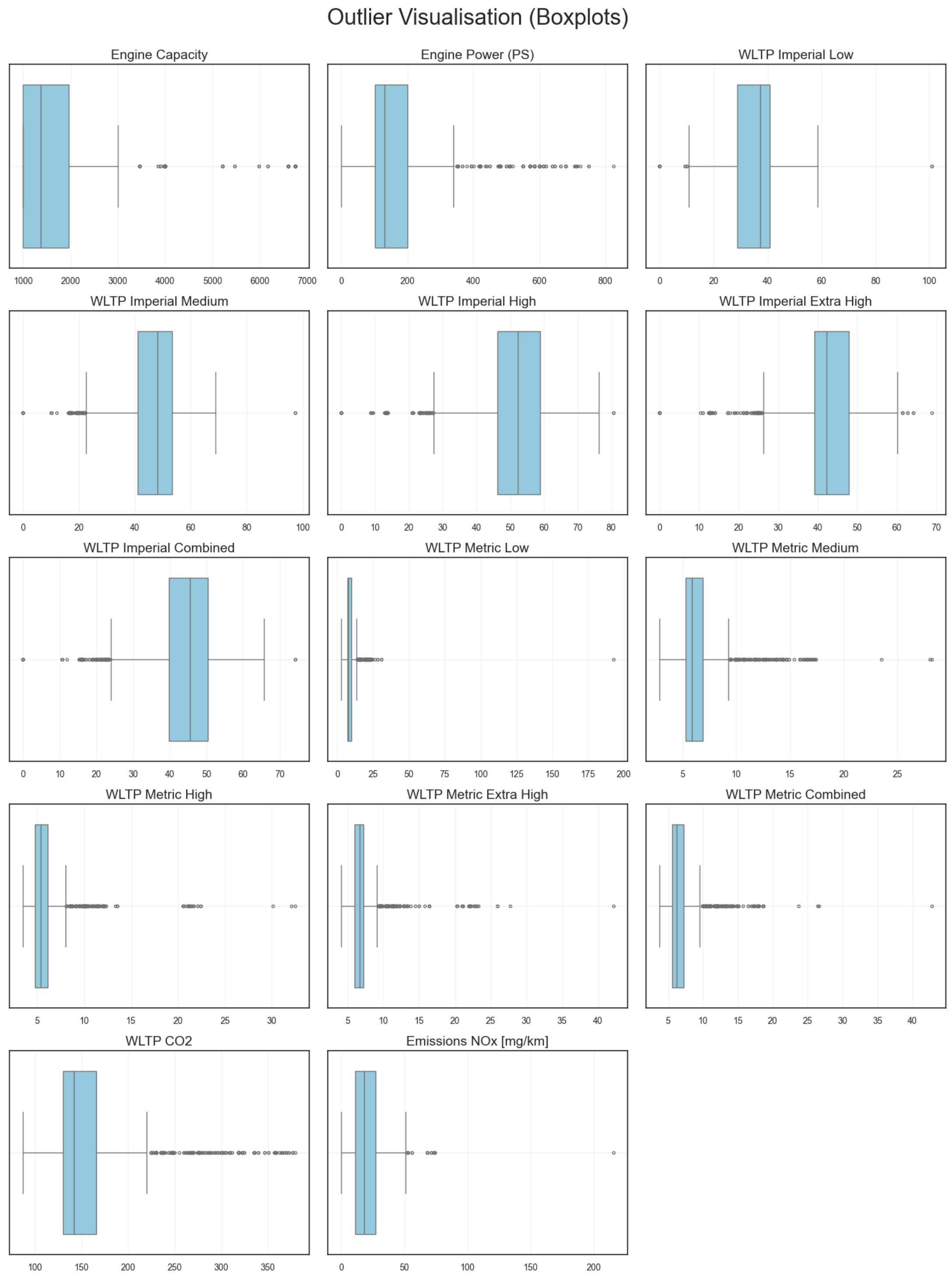


3.5 Outlier Check

To further improve our model and increase the accuracy, we have perform an analysis of the outliers for both datasets. In the fuel vehicles, we can clearly see the outliers are strongly skewed in most of the WLTP Metric variables and highly skewed in Engine Capacity and Power. This could be due to the high-performance models of vehicles. Since, the WLTP Imperial is having the same purpose with a different unit of measurements, we have decided to drop all the WLTP Metric variables which having higher outliers compare to MLTP Imperial. This table showing the updated model for *data_fuel*.

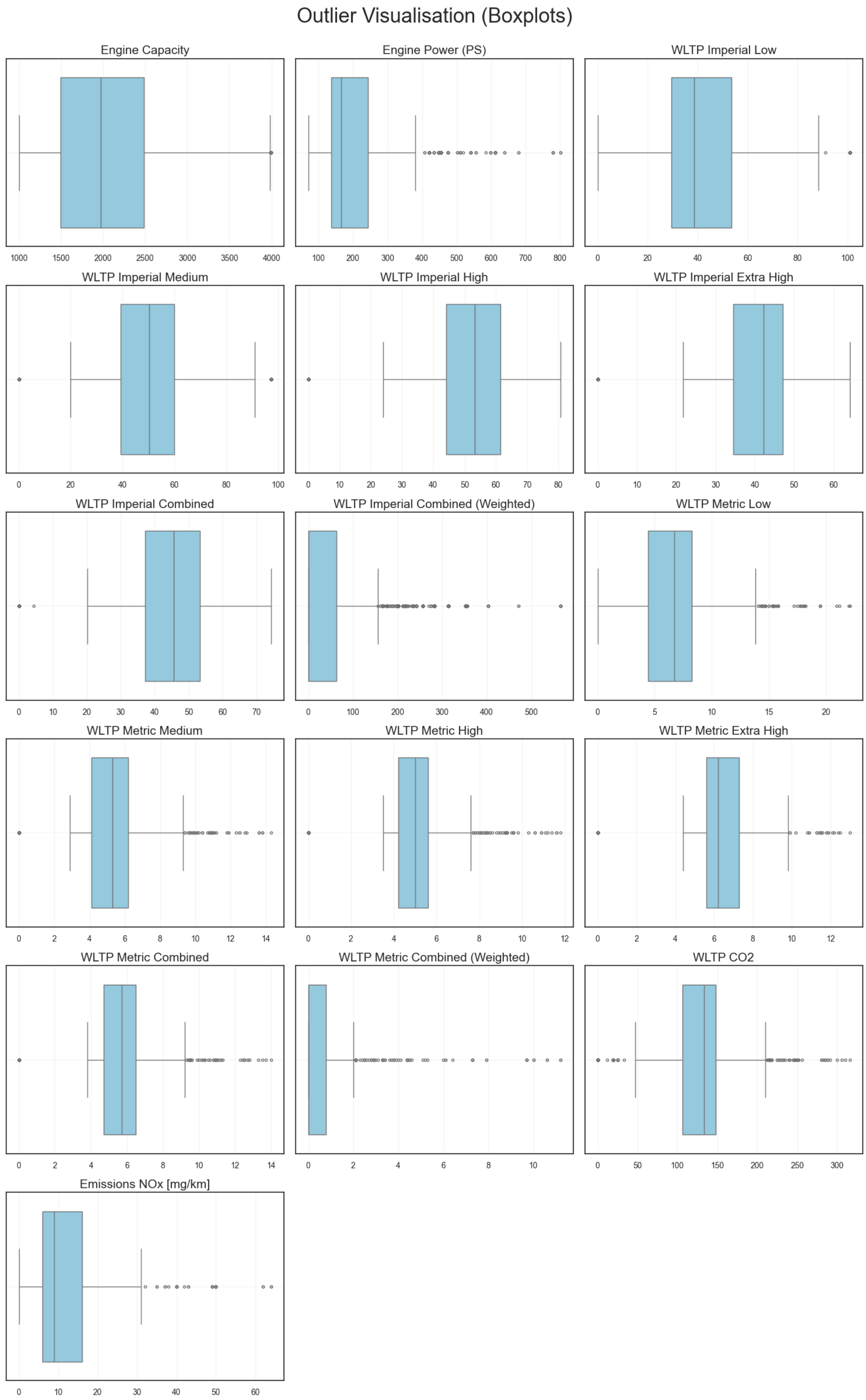
Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)

Category	Keep	Remove
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined	Diesel VED Supplement, Testing Scheme, WLTP Imperial Combined (Weighted), WLTP Metric Combined (Weighted), WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard		Euro Standard
Emmission	Emissions NOx [mg/km], WLTP CO2	WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change



Same for fuel vehicles, it seems to be a high outliers in WLTP Metric for hybrid vehicles. Therefore, these variables will be removed from hybrid vehicles dataset too. Here is the table showing the updated columns remain in *data_hybrid*.

Category	Keep	Remove
Vehicle	Manual or Automatic, Engine Capacity, Fuel Type, Powertrain, Engine Power (PS)	Manufacturer, Model, Description, Transmission, Engine Power (Kw)
Fuel Consumption	WLTP Imperial Low, WLTP Imperial Medium, WLTP Imperial High, WLTP Imperial Extra High, WLTP Imperial Combined	Diesel VED Supplement, Testing Scheme, WLTP Metric Low, WLTP Metric Medium, WLTP Metric High, WLTP Metric Extra High, WLTP Metric Combined, WLTP Metric Combined (Weighted), WLTP Imperial Combined (Weighted)
Electric Consumption		Electric energy consumption Miles/kWh, wh/km ,Maximum range (Km), Maximum range (Miles), Equivalent All Electric Range Miles, Equivalent All Electric Range KM, Electric Range City Miles, Electric Range City Km
Standard		Euro Standard
Emmission	Emissions NOx [mg/km], WLTP CO2	WLTP CO2 Weighted , THC Emissions [mg/km], THC + NOx Emissions [mg/km], Particulates [No.] [mg/km], RDE NOx Urban, RDE NOx Combined, Noise Level dB(A)
Date		Date of change


Outlier Visualisation (Boxplots)
Engine Capacity
1000 1500 2000 2500 3000 3500 4000
Engine Power (PS)
100 200 300 400 500 600 700 800
WLTP Imperial Low
0 20 40 60 80 100
WLTP Imperial Medium
0 20 40 60 80 100
WLTP Imperial High
0 10 20 30 40 50 60 70 80
WLTP Imperial Extra High
0 10 20 30 40 50 60
WLTP Imperial Combined
0 10 20 30 40 50 60 70
WLTP Imperial Combined (Weighted)
0 100 200 300 400 500
WLTP Metric Low
0 5 10 15 20
WLTP Metric Medium
0 2 4 6 8 10 12 14
WLTP Metric High
0 2 4 6 8 10 12
WLTP Metric Extra High
0 2 4 6 8 10 12
WLTP Metric Combined
0 2 4 6 8 10 12 14
WLTP Metric Combined (Weighted)
0 2 4 6 8 10
WLTP CO2
0 50 100 150 200 250 300
Emissions NOx [mg/km]
0 10 20 30 40 50 60

3.6 Cross Validation

Now, we have the final model with all the features and target variables to train and test and model. Before we start the model, a *data_split* function was created to split the *data_fuel* and *data_hybrid* into train, validate, and test datasets. The datasets were split into **80% for training, 10% for validations and 10% for testing**.

After splitting the data, we have created a function called *train_lasso_with_cv* to use **Lasso** model for feature selection with a **10-fold** cross validations. Based on the results from Lasso model, we have designed to refine the Lasso model again with the best L1 regularisation strength (*alphas*). With the results from Lasso model, we can understand the errors using **mean squared error (MSE)** and variances by calculating the **r2_score** as well as listing the coefficient of each features.

We have done the training for both datasets *data_fuel* and _data_hybrid with the two target variables *WLTP CO2* and *Emissions NOx [mg/km]* that we have defined above. Below table is the results from the *lasso* model training with *10-fold* cross validation.

Metric	Fuel / CO2 (12 features)	Fuel / CO2 (3 features)	Fuel / NOx (12 features)	Fuel / NOx (3 features)	Hybrid / CO2 (12 features)	Hybrid / CO2 (3 features)	Hybrid / NOx (12 features)	Hybrid / NOx (3 features)
Best alpha	0.054	0.022	0.022	0.002	0.103	0.022	0.004	0.055
Train RMSE	197.082	2071.979	213.583	238.598	1359.811	2071.979	45.390	74.931
Validate RMSE	270.483	1912.145	1272.888	171.254	1044.115	1912.145	71.701	98.815
Validate R^2	0.926	0.055	-6.459	-0.003	0.484	0.055	0.279	0.007
Test RMSE	247.201	2080.948	1172.173	149.236	1525.036	2080.948	24.205	45.544
Test R^2	0.890	0.251	6.831	0.002	0.451	0.251	0.463	-0.009

Based on the *alphas* returned from first round of training, we have run the *lasso* model again with following results.

Metric (refined)	Fuel / CO2 (12 features)	Fuel / CO2 (3 features)	Fuel / NOx (12 features)	Fuel / NOx (3 features)	Hybrid / CO2 (12 features)	Hybrid / CO2 (3 features)	Hybrid / NOx (12 features)	Hybrid / NOx (3 features)
Validate RMSE	261.047	1912.280	1276.104	171.228	1043.926	1912.280	71.691	98.813
Validate R^2	0.928	0.055	-6.478	-0.003	0.484	0.055	0.279	0.007
Test RMSE	236.887	2081.085	1175.241	149.221	1525.295	2081.085	24.225	45.544
Test R^2	0.894	0.251	-6.852	0.003	0.451	0.251	0.462	-0.009

From the *lasso* training, we found that the model with all the 12 features are working fine in both datasets for fuel and hybrid vehicle when we examine the correlation with CO2 and NOx. The errors have increased drastically after reduced to 3 features only. Eventhough the model for Fuel/NOx was strongly negative in validation R2, we are keeping the 12 features to standardised the features selection. We will consider an algorithm improvements in the future.

Next, we have done a diagnostic using **residual/prediction scatter plot** to check the scalar metrics on the 12 features.

3.6.1 Fuel / WLTP CO2

The result showing the residual scatter is tight, centered on zero, and roughly uniform across fitted values. The histogram is symmetric with light tails that matches the small residual mean and standard deviation (~0, 14) and strong validation R^2 (~0.93), so the model is well calibrated with no obvious bias or variances.

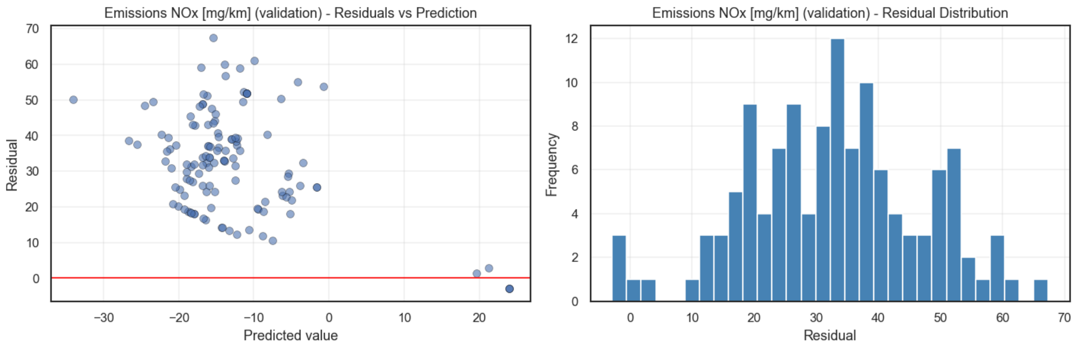
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Fuel / WLTP CO2 (12 features)	Lasso	0.054	14.039	16.446	0.926	15.723	0.890
Fuel / WLTP CO2 (12 features)	Ridge	2.976	13.994	24.107	0.840	24.144	0.740

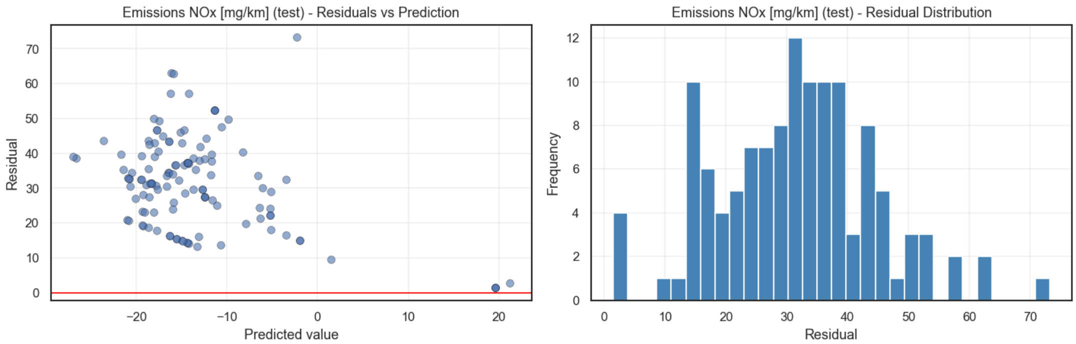
With the reduced 3-feature set, the result from *Lasso* matches the result *Ridge* model. That tells us the simplified feature gives lower error with less tuning sensitivity, it's the stronger feature-selection choice for fuel and CO2 mode.

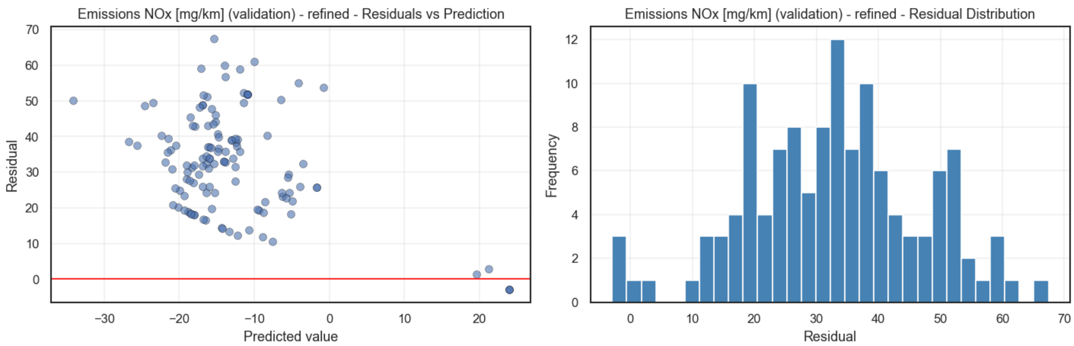
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Fuel / WLTP CO2 (3 features)	Lasso	0.143	14.786	14.781	0.940	13.299	0.921
Fuel / WLTP CO2 (3 features)	Ridge	2.976	14.787	14.790	0.940	13.343	0.921

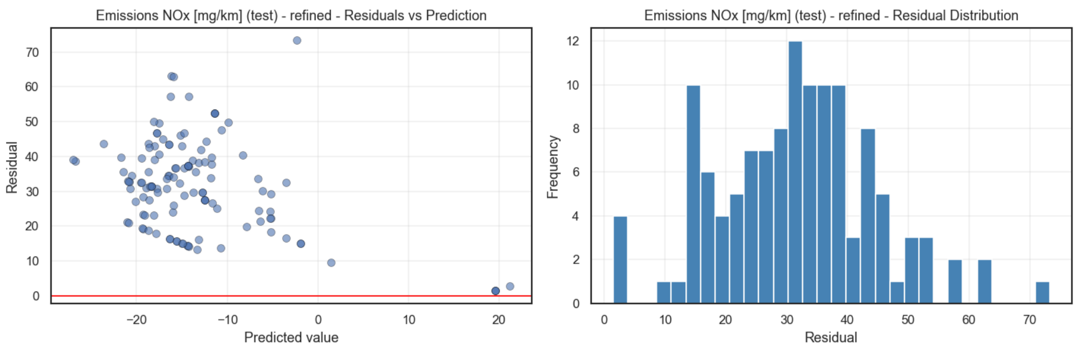
3.6.2 Fuel / NOx

In fuel / NOx model, the residuals are mostly located above zero. The model seems to be under-predicts high NOx readings, which is why validation/test R^2 plunge well below zero. The histogram's heavy right tail confirms the bias and we might need to add more features or change a different model for fuel vehicles and NOx emission prediction.









Both results from *Lasso* and *Ridge* tells the 12-features models give negative value in R2 and RMSE jump to around 35.

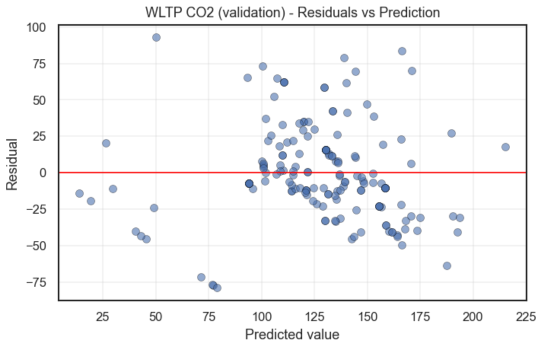
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Fuel / NOx (12 features)	Lasso	0.023	14.614	35.678	-6.460	34.237	-6.832
Fuel / NOx (12 features)	Ridge	2.976	14.613	36.945	-6.999	35.501	-7.421

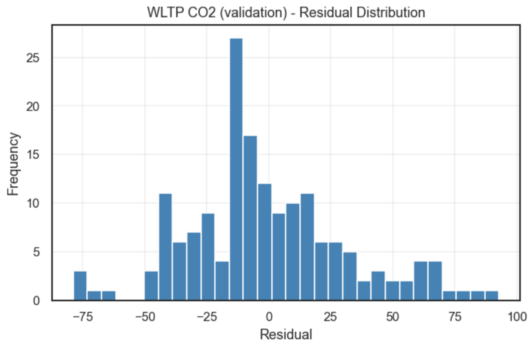
Whereas in 3-features models for fuel / NOx is more stable with low RMSE but it is heavy regularised in *Ridge* with a high *alpha* at 234. In conclusion, the features we have selected do not fit well in fuel and NOx prediction.

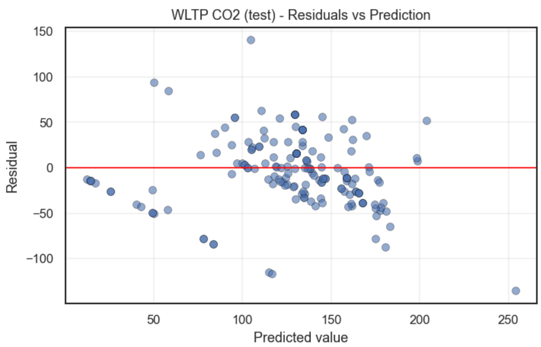
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Fuel / NOx (3 features)	Lasso	0.002	15.447	13.086	-0.004	12.216	0.003
Fuel / NOx (3 features)	Ridge	233.572	15.479	12.929	0.020	12.133	0.016

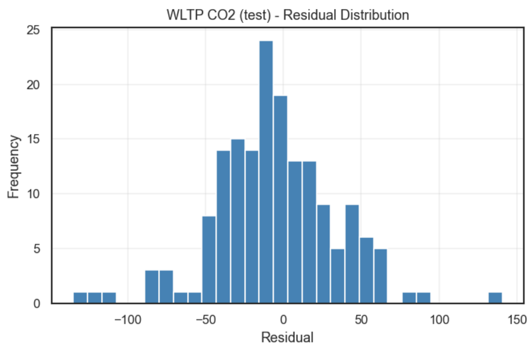
3.6.3 Hybrid / CO2

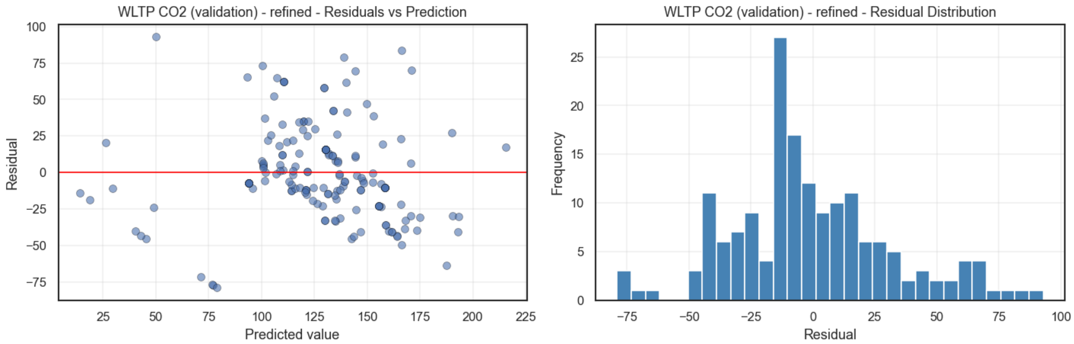
Residual cloud is symmetric but very wide (+/-140), showing high variance even though it remains centered. That matches RMSE ~32–39 and explains only ~45–48% of variance, we might consider adding more features or a nonlinear terms to lower down the variance.

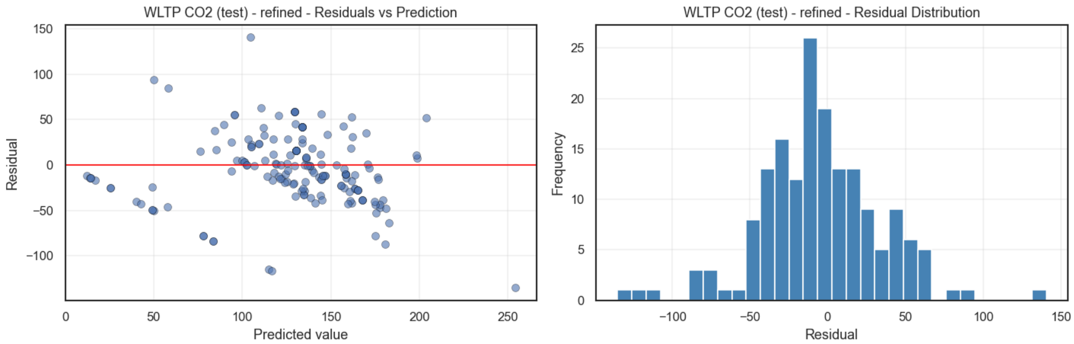












When comparing the 12-features set for hybrid / CO2 model, the results from *Lasso* and *Ridge* are about the same.

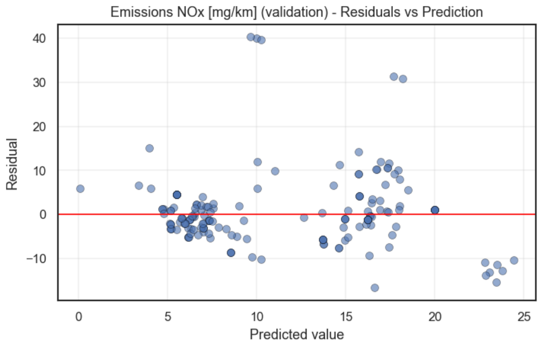
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Hybrid / CO2 (12 features)	Lasso	0.103	36.876	32.313	0.484	39.052	0.451
Hybrid / CO2 (12 features)	Ridge	6.158	36.881	32.256	0.486	39.147	0.448

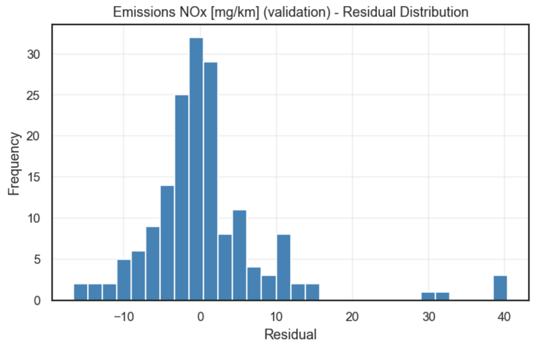
When reducing to 3-features set, the accuracy has dropped sharply. Hence, we can conclude that 12-features set is a better model in predicting hybrid / CO2.

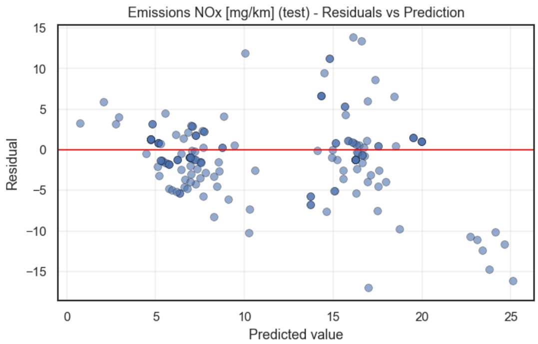
Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Hyrid / CO2 (3 features)	Lasso	0.022	45.519	43.728	0.055	45.617	0.251
Hyrid / CO2 (3 features)	Ridge	6.158	45.520	43.691	0.057	45.595	0.252

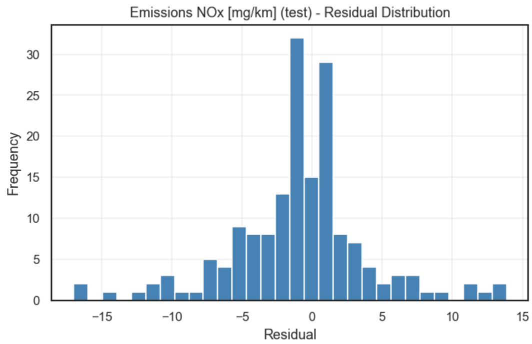
3.6.4 Hybrid / NOx

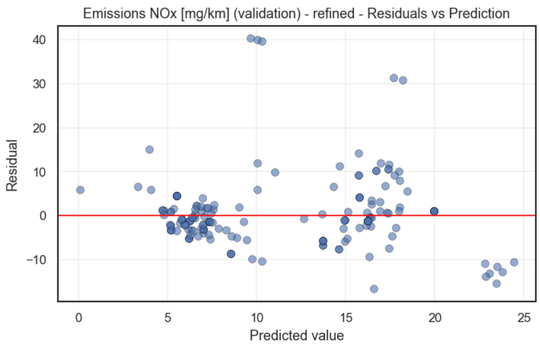
Residuals are compact, centered, and evenly spread, giving a clean plot with no obvious bias. Validation/test R2 around 0.28–0.46 confirm a solid fit.

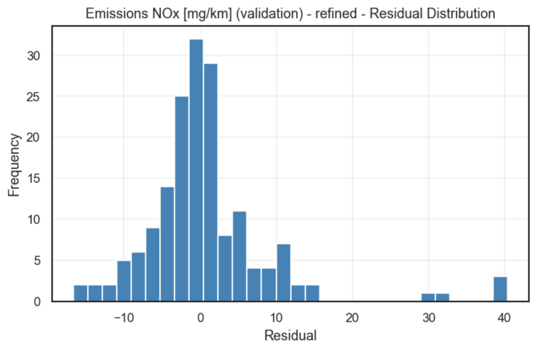

Emissions NOx [mg/km] (validation) - Residuals vs Prediction
Residual
Predicted value
40
30
20
10
0
−10
0
5
10
15
20
25

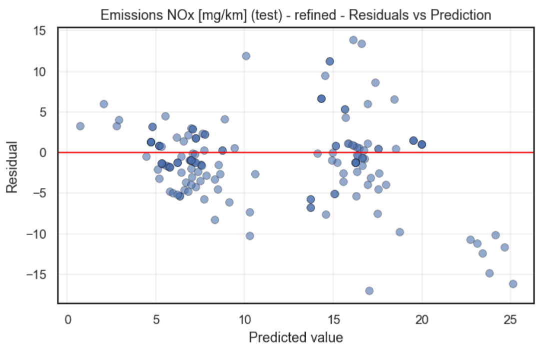

Emissions NOx [mg/km] (validation) - Residual Distribution
Frequency
Residual
30
25
20
15
10
5
0
−10
0
10
20
30
40

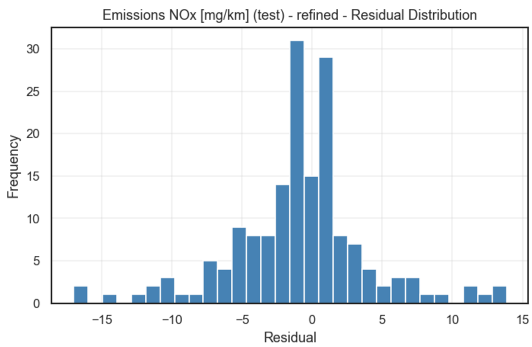

Emissions NOx [mg/km] (test) - Residuals vs Prediction
Residual
Predicted value
15
10
5
0
−5
−10
−15
0
5
10
15
20
25


Emissions NOx [mg/km] (test) - Residual Distribution
Frequency
Residual
30
25
20
15
10
5
0
−15
−10
−5
0
5
10
15


Emissions NOx [mg/km] (validation) - refined - Residuals vs Prediction
Residual
Predicted value
40
30
20
10
0
−10
0
5
10
15
20
25


Emissions NOx [mg/km] (validation) - refined - Residual Distribution
Frequency
Residual
30
25
20
15
10
5
0
−10
0
10
20
30
40


Emissions NOx [mg/km] (test) - refined - Residuals vs Prediction
Residual
Predicted value
15
10
5
0
−5
−10
−15
0
5
10
15
20
25


Emissions NOx [mg/km] (test) - refined - Residual Distribution
Frequency
Residual
30
25
20
15
10
5
0
−15
−10
−5
0
5
10
15

For hybrid / NOx prediction, the 12-features model are doing great in both validation and testing but the *Ridge* model has chosen a high *alpha* at 26.4 which could be due to the noisy target variables or small hybrid datasets.

Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Hyrid / NOx (12 features)	Lasso	0.005	6.737	8.468	0.279	4.920	0.463
Hyrid / NOx (12 features)	Ridge	26.367	6.752	8.481	0.277	4.805	0.488

The 3-features model for hybrid / NOx prediction has returned high errors in both *Lasso* and *Ridge* models. Hence, we will not consider this model.

Label	Variant	Best	Train RMSE	Val RMSE	Val R^2	Test RMSE	Test R^2
Hyrid / NOx (3 features)	Lasso	0.056	8.656	9.941	0.007	6.749	-0.010
Hyrid / NOx (3 features)	Ridge	1000.000	8.658	9.968	0.002	6.746	-0.009

Across the regularised experiments, 12-features model are performing better than 3-features model, with higher accuracy and less variants. The fuel/CO2 emerges as the most predictable target, follow by hybrid CO2 and NOx. Residual diagnostics therefore suggest targeted feature engineering or alternative models for the fuel/NOx task, where even the best configuration retains a biased right tail.

4 Neural Network - Regression Tasks

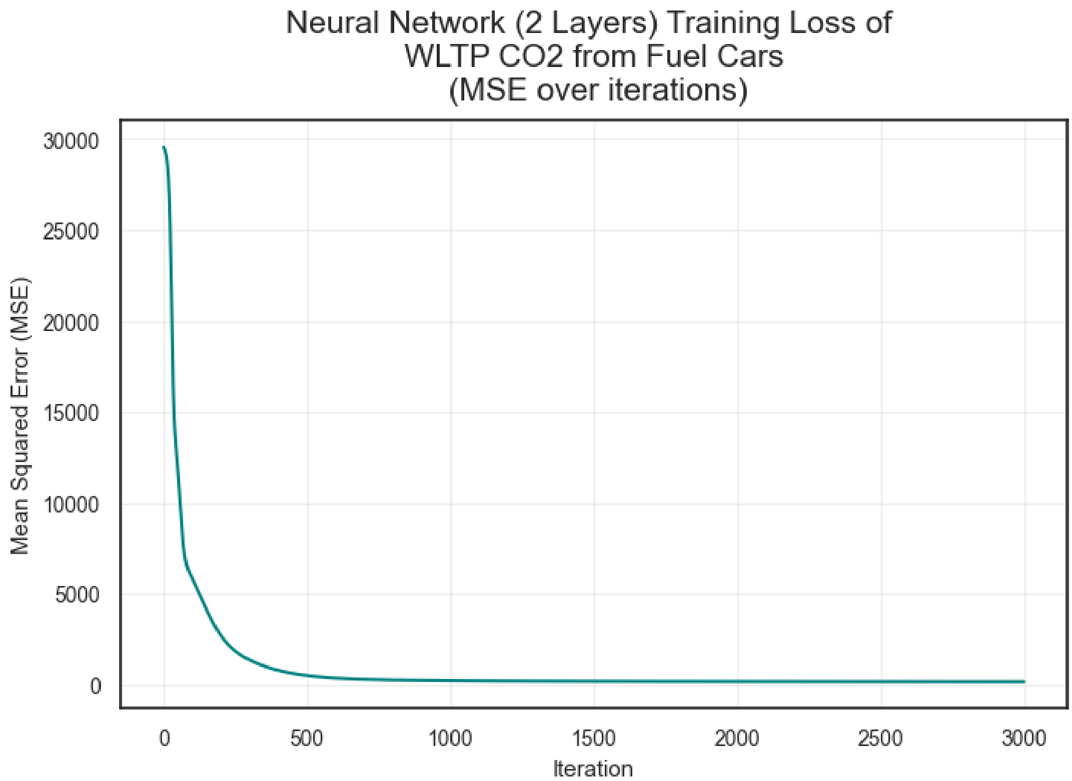
Variables Selection: All of the remaining variables after data preprocessing will be used for neural network regression models. Before bulding the models, examining the current features and data shape in the two datasets will be helpful to decide the relevant number of nodes in hidden layers, step size, and iterations.

Model Description As this is a regression task and both of target variables are numeric, hidden layers will be applied with ReLU - max(0,x) as activation method to keep gradients strong and captures nonlinearity. The output layer will be applied with linear approach without activation to allow continuous outputs from the both gas emission types.

Parameters to train all the model: * Step size: 0.002. This is considered as relevant step size since the two datasets are small (< 5,000 rows). This step size also ensure sufficient learning for models when training. Small values lower than 0.0005 can cause overfitting for the models depending on current data scale. * Iteration: 3,001 (). Since this is a small datatset, iteration times above 2,000 will be relevant for the model training. * Number of nodes in each layer: 10-60 nodes. Given the current features of two datasets are 12, the number of nodes for each hidden layer should range from 6 (0.5x numbers of features) to 60 (3x numbers of features) nodes.

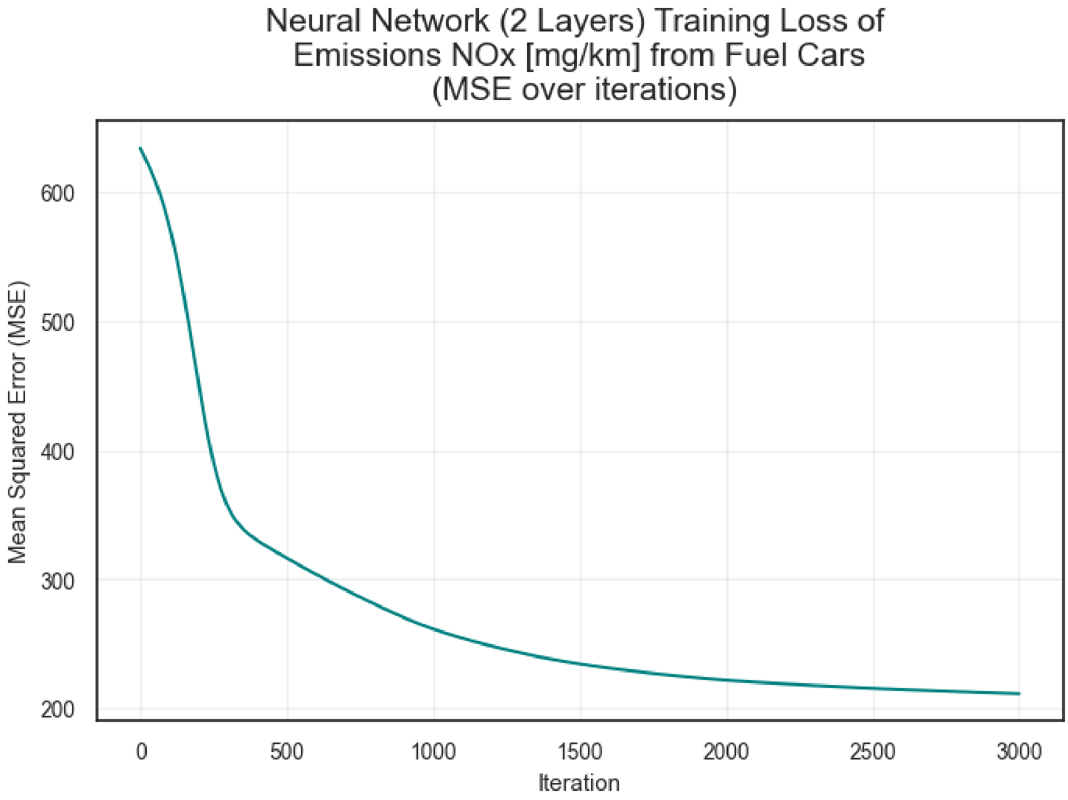
4.1 Neural Network Regression (2 Layers)

Metric	Value
Training RMSE	0.007827
Coefficient of determination (R^2) on training data	1.000000
Test RMSE	69.798494
Coefficient of determination (R^2) on test data	-1.171900
Validation RMSE	72.883765
Coefficient of determination (R^2) on validation data	-0.459300



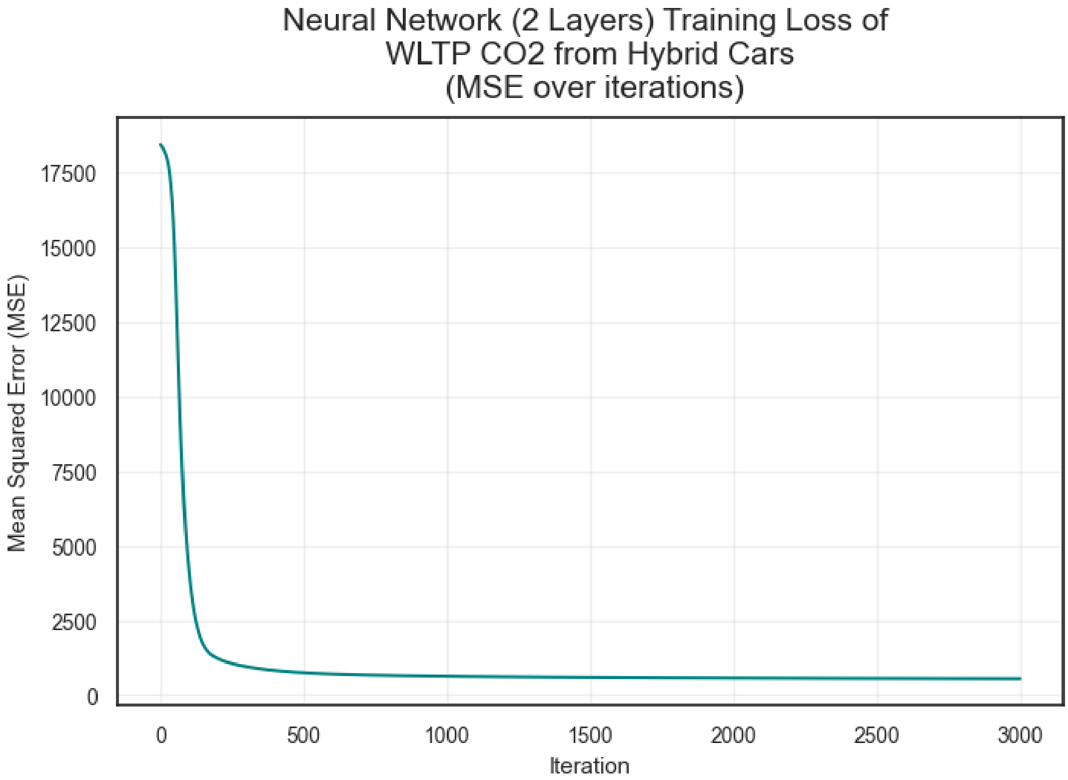
The neural network model (2 layers) used for predicting CO2 emissions from fuel cars seem to be overfitting due to high accuracy on training set but significantly low accuracy in both test and validation set. Therefore, this model will be removed.

Metric	Value
Training RMSE	0.001051
Coefficient of determination (R^2) on training data	1.000000
Test RMSE	14.862229
Coefficient of determination (R^2) on test data	-0.475800
Validation RMSE	16.466513
Coefficient of determination (R^2) on validation data	-0.589100



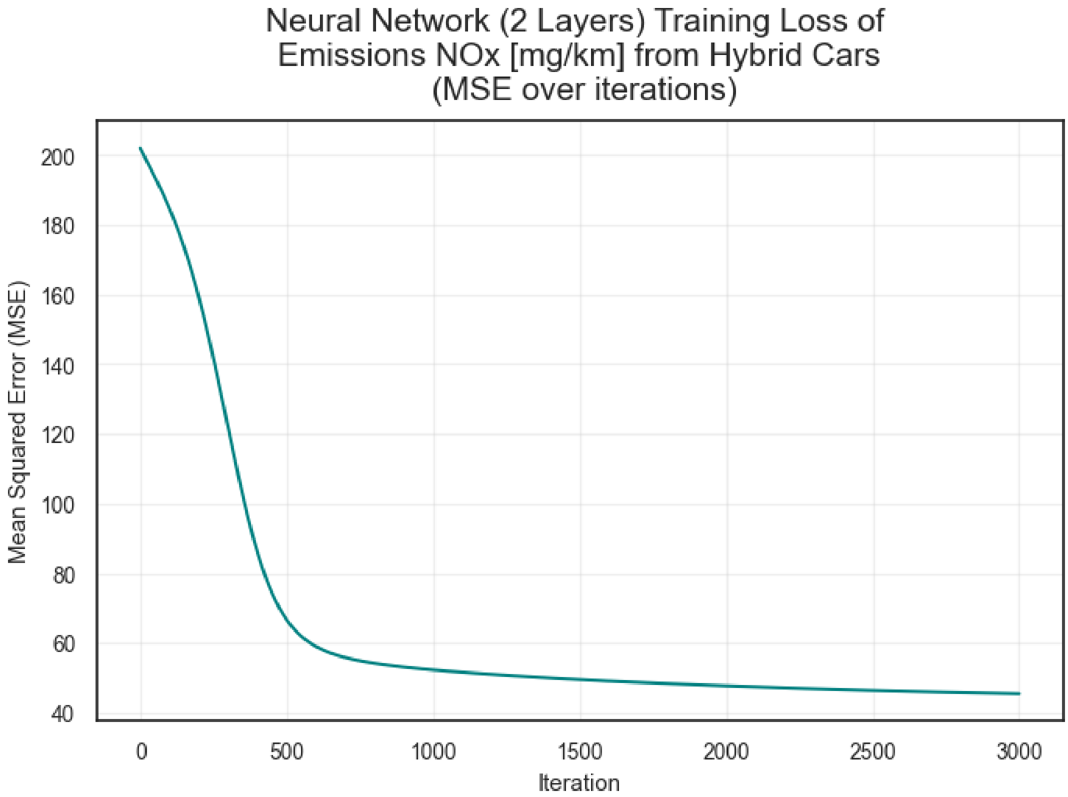
The neural network model (2 layers) used for predicting NOx emissions from fuel cars seem to be overfitting due to high accuracy on training set but significantly low accuracy in both test and validation set. Therefore, this model will be removed.

Metric	Value
Training RMSE	0.004785
Coefficient of determination (R^2) on training data	1.000000
Test RMSE	27.859055
Coefficient of determination (R^2) on test data	0.720600
Validation RMSE	18.148563
Coefficient of determination (R^2) on validation data	0.837300



The neural network model (2 layers) used for predicting CO2 emissions from hybrid cars has lower gaps of accuracy and RMSE between training, and test and validation set. Thus, it can be a potential model for further evaluation as this stage.

Metric	Value
Training RMSE	0.000539
Coefficient of determination (R^2) on training data	1.000000
Test RMSE	4.646769
Coefficient of determination (R^2) on test data	0.521200
Validation RMSE	8.297508
Coefficient of determination (R^2) on validation data	0.308100



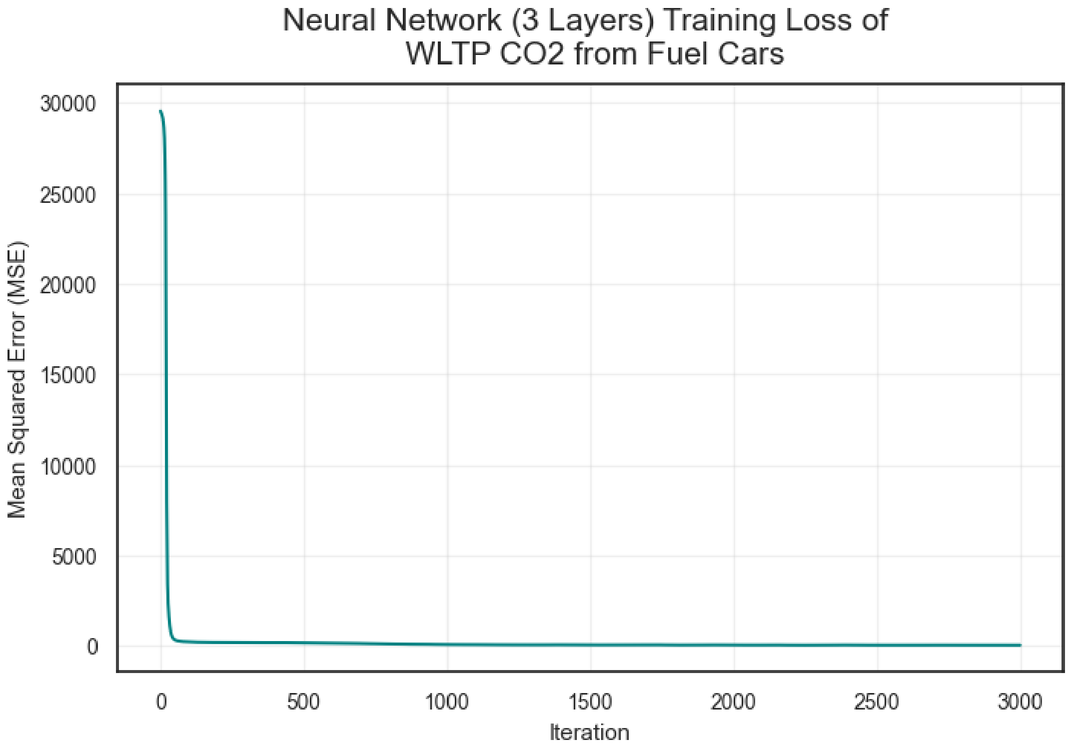
The neural network model (2 layers) used for predicting NOx emissions from hybrid cars seem to be overfitting due to high accuracy on training set but low accuracy in both test and validation set. Therefore, this model will be removed.

Findings: Only 1 NN model can be selected at this stage which is for CO2 prediction of hybrid cars. Since the current models have only 2 layers which is quite simple, leading to biased results, other models should be built with 1 more layer for comprehensive comparisons and selection, leading to Neural Network Regression (3 Layers).

4.2 Neural Network Regression - 3 Layers

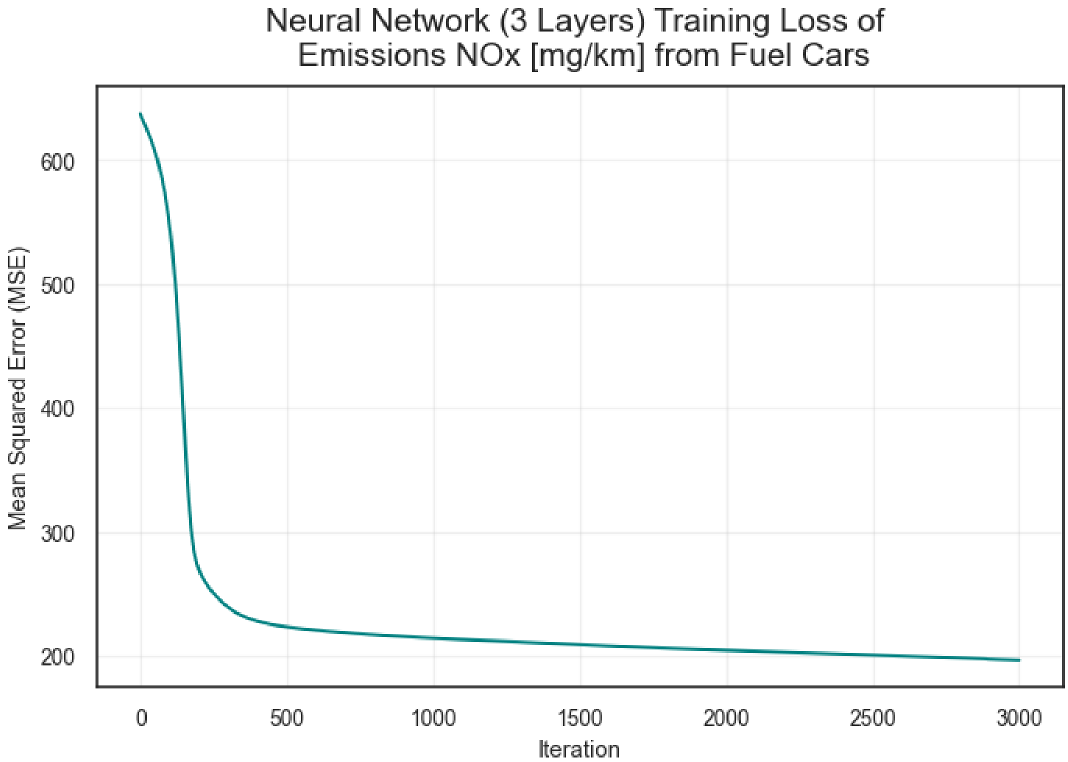
In order to to achieve relevant model comparison, all of the current parameters in step size, and iterations will be kept. The first layer still have 60 nodes similar to the previous models. However, the second layer will have 30 nodes.

Metric	Value
Training RMSE	5.726886
Coefficient of determination (R^2) on training data	0.989200
Test RMSE	61.253981
Coefficient of determination (R^2) on test data	-0.672700
Validation RMSE	61.625088
Coefficient of determination (R^2) on validation data	-0.043300



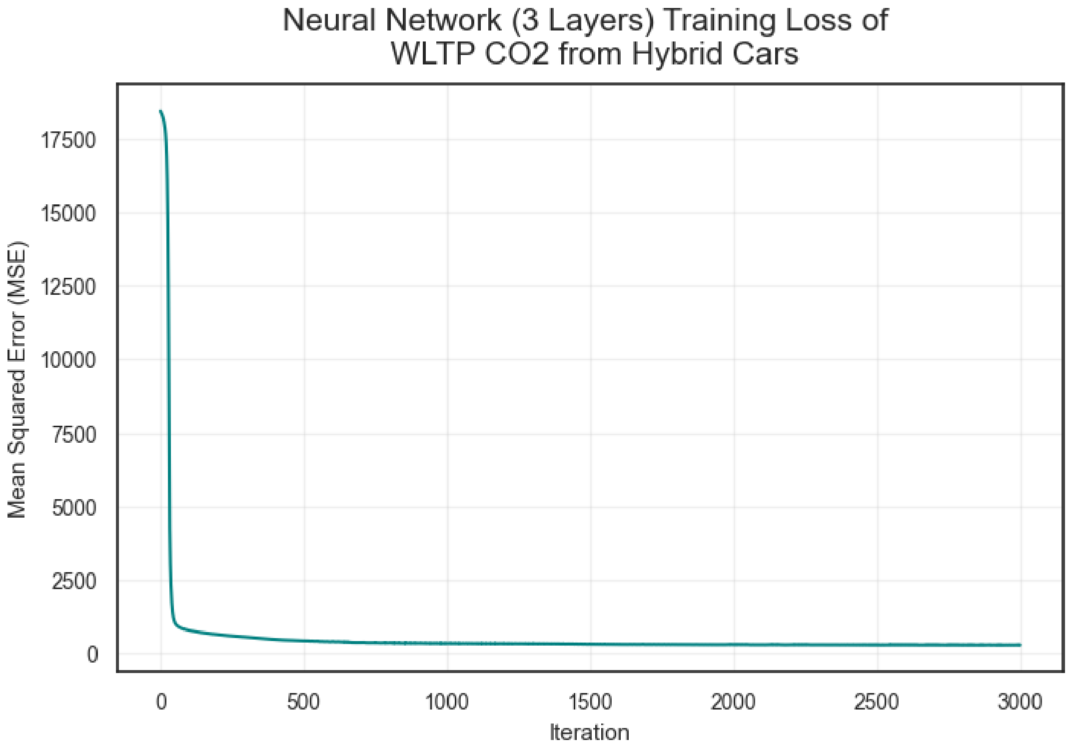
The neural network model (3 layers) used for predicting CO2 emissions from fuel cars seem to be overfitting again due to high accuracy on training set but significantly low accuracy in both test and validation set. Therefore, this model will be removed.

Metric	Value
Training RMSE	14.015439
Coefficient of determination (R^2) on training data	0.199100
Test RMSE	16.430206
Coefficient of determination (R^2) on test data	-0.803600
Validation RMSE	18.211237
Coefficient of determination (R^2) on validation data	-0.943700



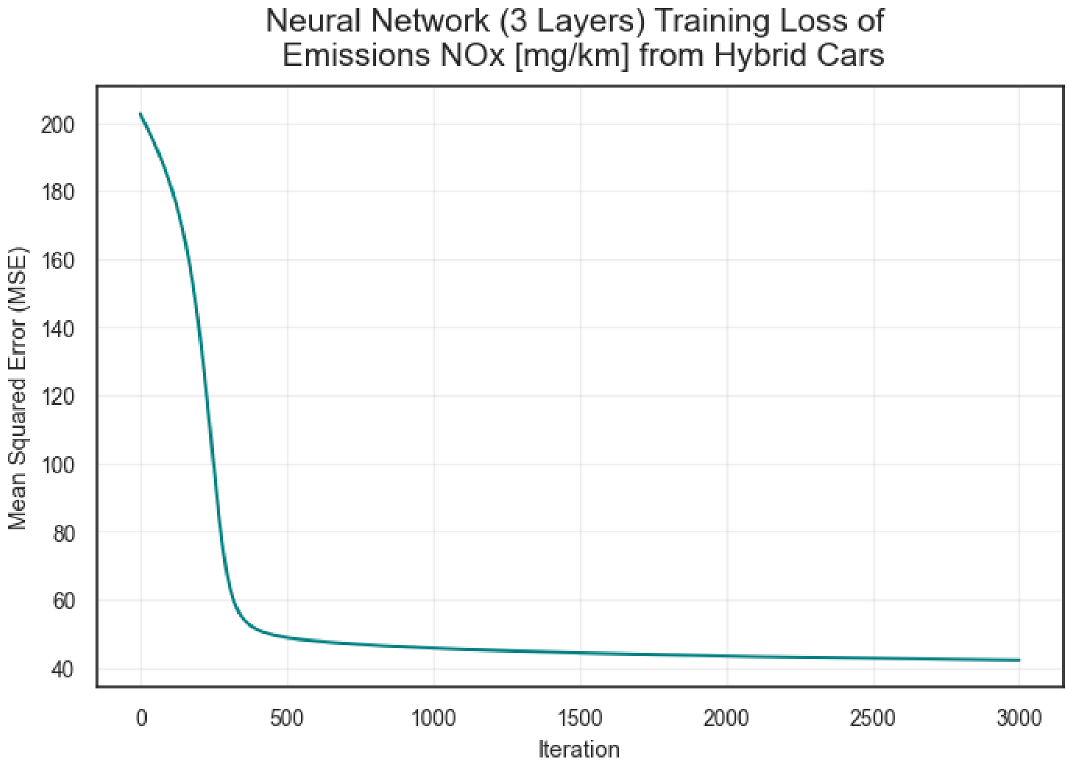
The neural network model (3 layers) used for predicting NOx emissions from fuel cars seem to be overfitting again due to high accuracy on training set but significantly low accuracy in both test and validation set. Therefore, this model will be removed.

Metric	Value
Training RMSE	16.844756
Coefficient of determination (R^2) on training data	0.903200
Test RMSE	16.965880
Coefficient of determination (R^2) on test data	0.896400
Validation RMSE	11.594324
Coefficient of determination (R^2) on validation data	0.933600



The neural network model (3 layers) used for predicting CO2 emissions from hybrid cars is a **potential model** due to having low differences of RMSE results between training and test set while having reasonable accuracy rate. Thus, this model will be kept for further evaluation on the next stage

Metric	Value
Training RMSE	6.504474
Coefficient of determination (R^2) on training data	0.437900
Test RMSE	4.777774
Coefficient of determination (R^2) on test data	0.493800
Validation RMSE	8.237239
Coefficient of determination (R^2) on validation data	0.318200



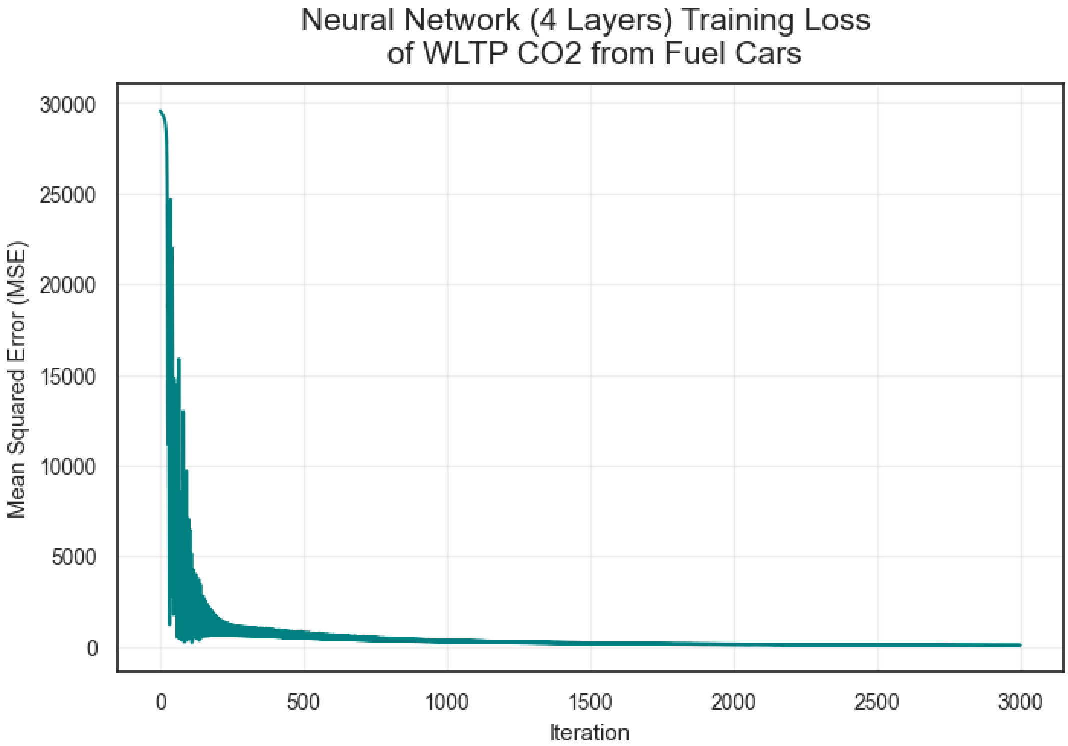
Despite having low gaps of accuracy and RMSE across different datasets, the accuracy score is quite low (nearly 50%). Thus, this model is not an effective model and should be removed at this stage

Findings: Similar to above findings, the similar model for CO2 predictions of hybrid car outperform other models within the same category. Thus, this model will be kepts for further comparison at the later stage Since there are no good models for predicting gas emissions (both types) for fuel cars, Neural Network Regression (4 Layers) should be built for further evaluation.

4.3 Neural Network Regression - 4 Layers

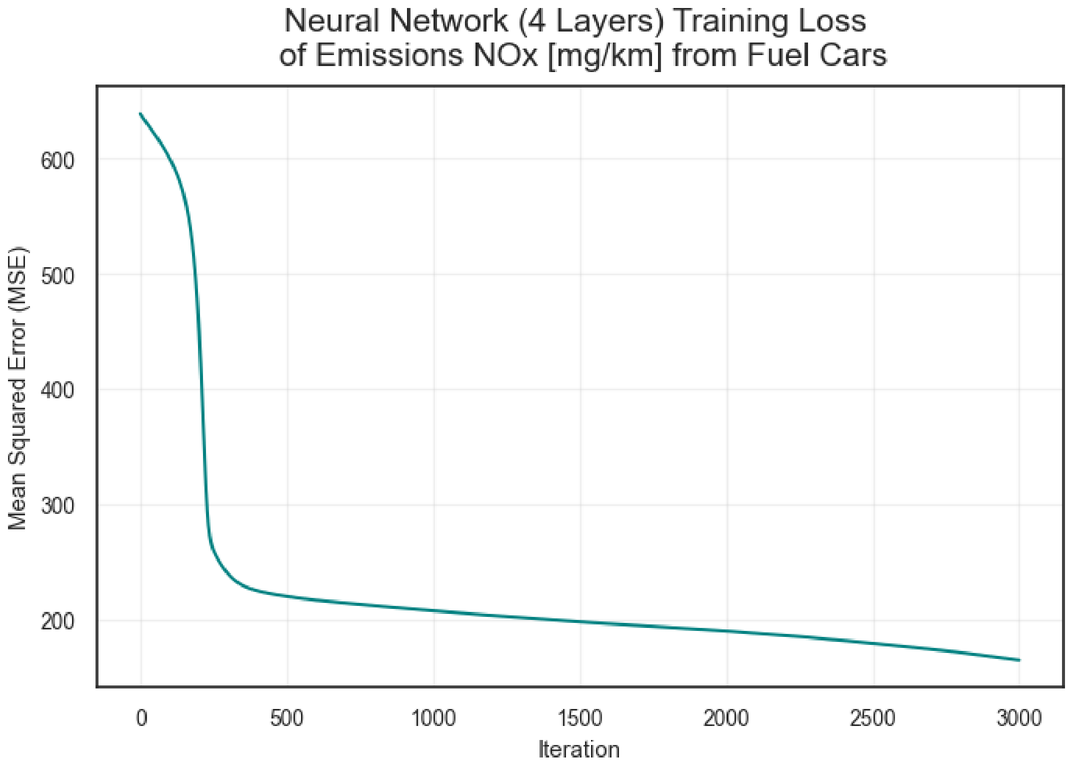
In order to to achieve relevant model comparison, all of the current parameters in step size, and iterations will be kept. The first layer and second layer still have similar numbers of nodes with the previous models, 60 and 30 respectively. There will be an additional third layer having 20 nodes.

Metric	Value
Training RMSE	8.069100
Coefficient of determination (R^2) on training data	0.978700
Test RMSE	23.891812
Coefficient of determination (R^2) on test data	0.745500
Validation RMSE	24.803494
Coefficient of determination (R^2) on validation data	0.831000



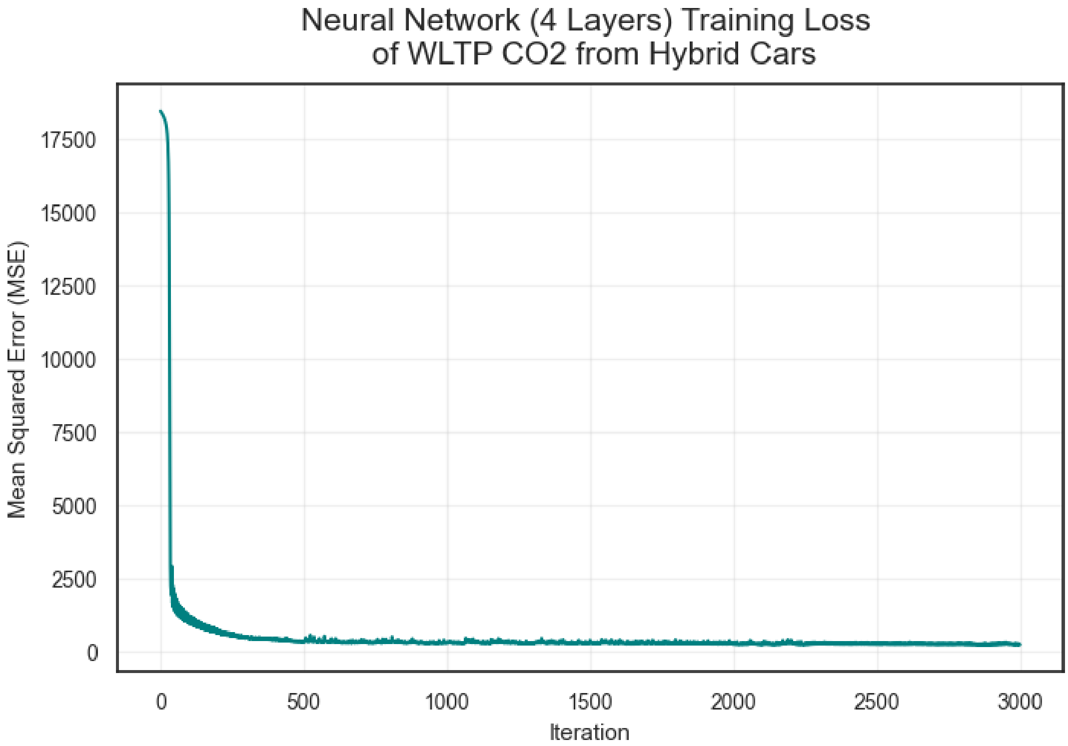
The neural network model (4 layers) used for predicting CO_2 emissions from fuel cars is a **potential model** as well due to having acceptable differences of RMSE results between training and test set. The accuracy rates across different sets are also acceptable with not very significant gaps. Thus, this model will be kept for further evaluation on the next stage.

Metric	Value
Training RMSE	12.853883
Coefficient of determination (R^2) on training data	0.326300
Test RMSE	18.305319
Coefficient of determination (R^2) on test data	-1.238800
Validation RMSE	19.945129
Coefficient of determination (R^2) on validation data	-1.331400



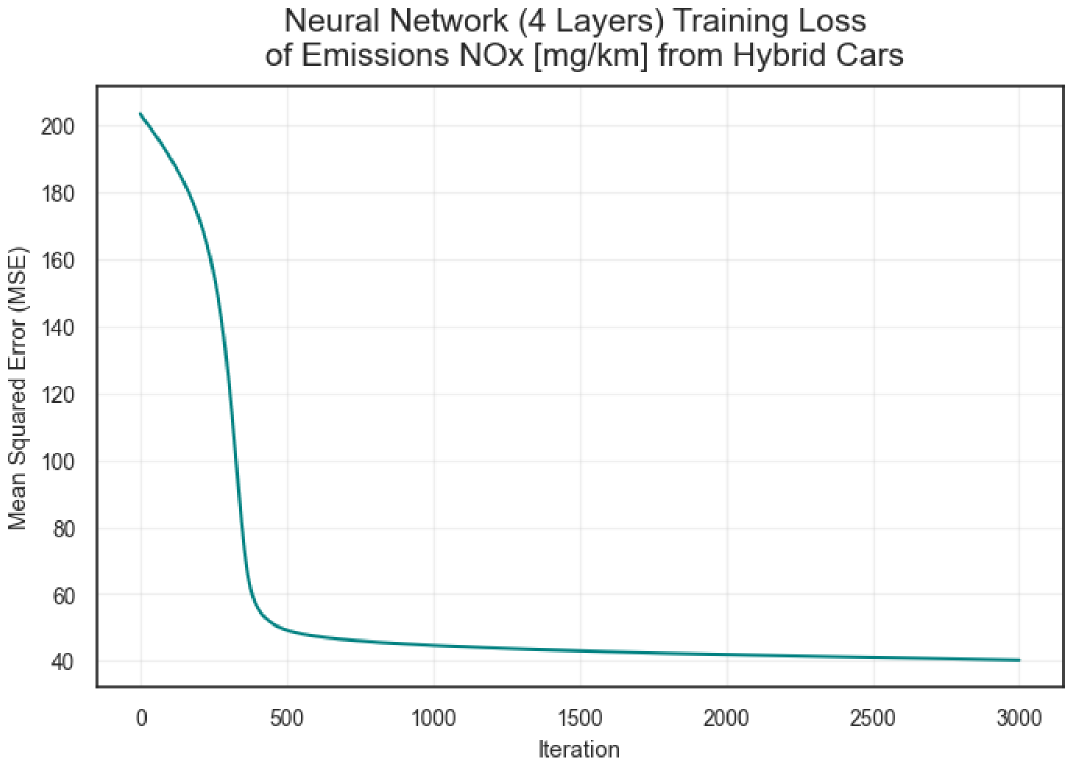
The neural network model (4 layers) used for predicting NOx emissions from fuel cars seem to be overfitting again due to similar patterns - the significant gaps of RMSE and accuracy rate across different datasets. Therefore, this model will be removed.

Metric	Value
Training RMSE	16.126670
Coefficient of determination (R^2) on training data	0.911300
Test RMSE	15.388180
Coefficient of determination (R^2) on test data	0.914800
Validation RMSE	11.271801
Coefficient of determination (R^2) on validation data	0.937200



Similar to above results of hybrid cars using CO2 emissions as the target variable, the neural network model (4 layers) used for predicting CO2 emissions from hybrid cars is a **potential model** due to having low differences of RMSE results between training and test set. The model also has relatively high and consistent accuracy rate across different datasets. Thus, this model will be kept for further evaluation on the next stage.

Metric	Value
Training RMSE	6.344130
Coefficient of determination (R^2) on training data	0.465300
Test RMSE	4.655104
Coefficient of determination (R^2) on test data	0.519500
Validation RMSE	8.134899
Coefficient of determination (R^2) on validation data	0.335000



There is also the similar patterns with the given hybrid car dataset and same target variable - NOx emissions, the accuracy score is quite low (below 50%). Thus, this model should be not be considered.

Findings from 4 Layers: There is an optimal regression model for predicting CO2 emissions of fuel cars now. Combining with above findings, CO2 prediction of hybrid cars tend to generate high and consistent results across different model complexity (number of layers). The below table will summarise the model selection process:

Summary of Neural Network Models for Fuel and Hybrid Cars

[63]:

	NN (2 Layers)	NN (3 Layers)	NN (4 Layers)
Fuel Cars, CO Emissions	No	No	Yes
Fuel Cars, NO Emissions	No	No	No
Hybrid Cars, CO Emissions	Yes	Yes	Yes
Hybrid Cars, NO Emissions	No	No	No

4.4 Model Selection

This stage is to select optimal models. These are the following selection criteria:
- Model accuracy (R2 Score).
- Performance consistency (the gaps between RMSE across different datasets).
- Model Complexity (the number of layers within the model).

CO2 predictions of Fuel Cars Given there is one model across different layers in this caterogy, neural network model with 4 layers is the selected model to predict CO2 of fuel cars

CO2 Predictions of Hybrid Cars Despite being selected in all model categories, the neural network (NN) model (3 layers) will be selected to predict CO2 of hybrid cars. Although the model (4 layers) have higher accuracy, the rate is just slightly higher whereas having higher complexity (additional layer). Along with that, the neural network model (2 layers) has relatively considerable gaps of both accuracy and RMSE in different datasets. Thus, the NN model (3 layers) is the optimal model in this category.

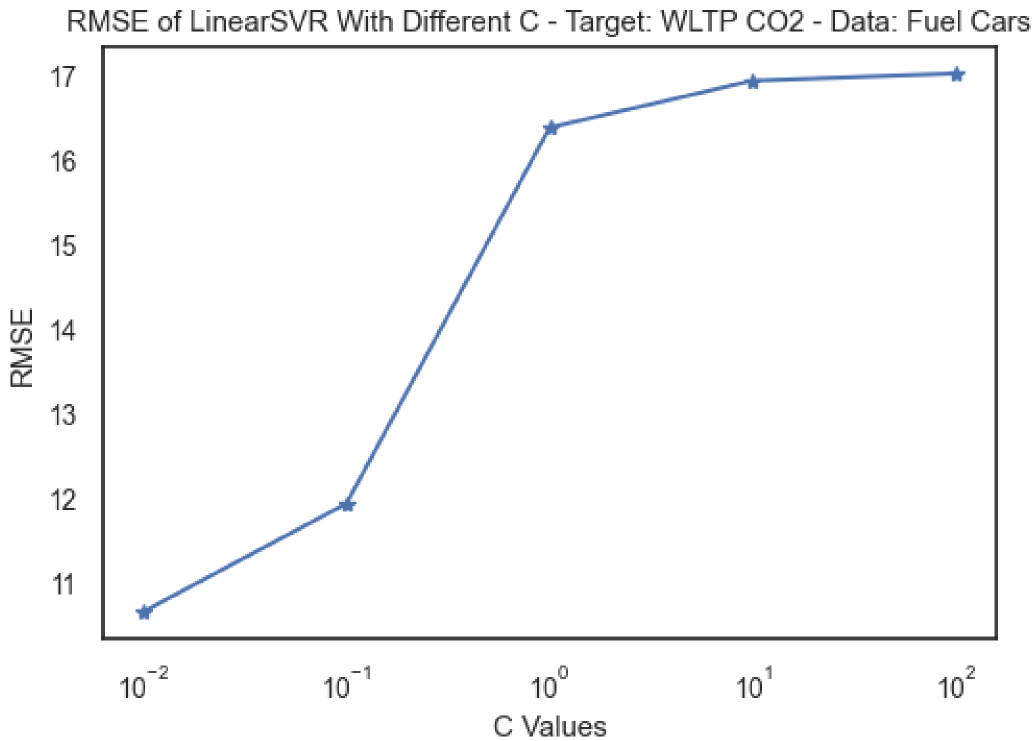
5 Support Vector Machine - Regression

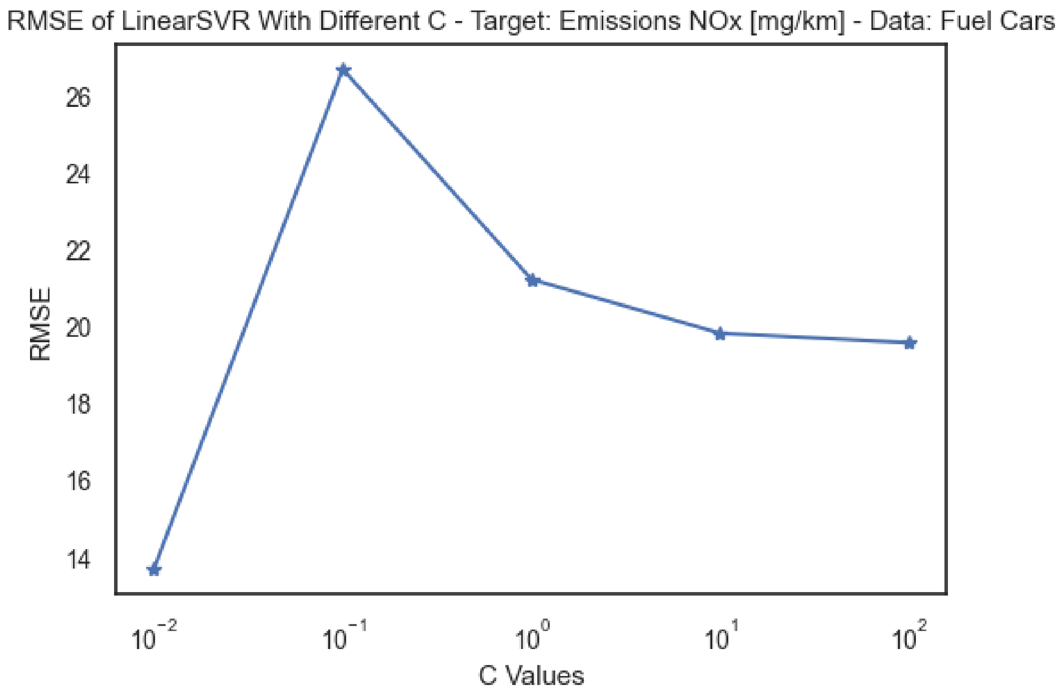
Approach

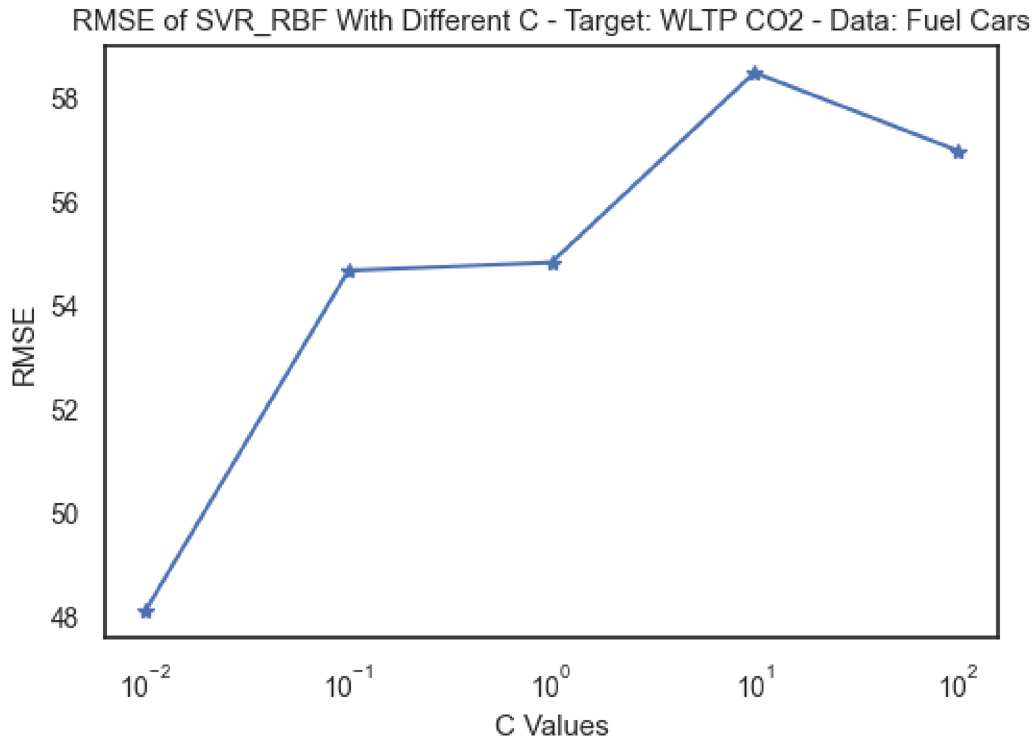
Step1: In each SVM model, a CV is conducted to find the best C margin.

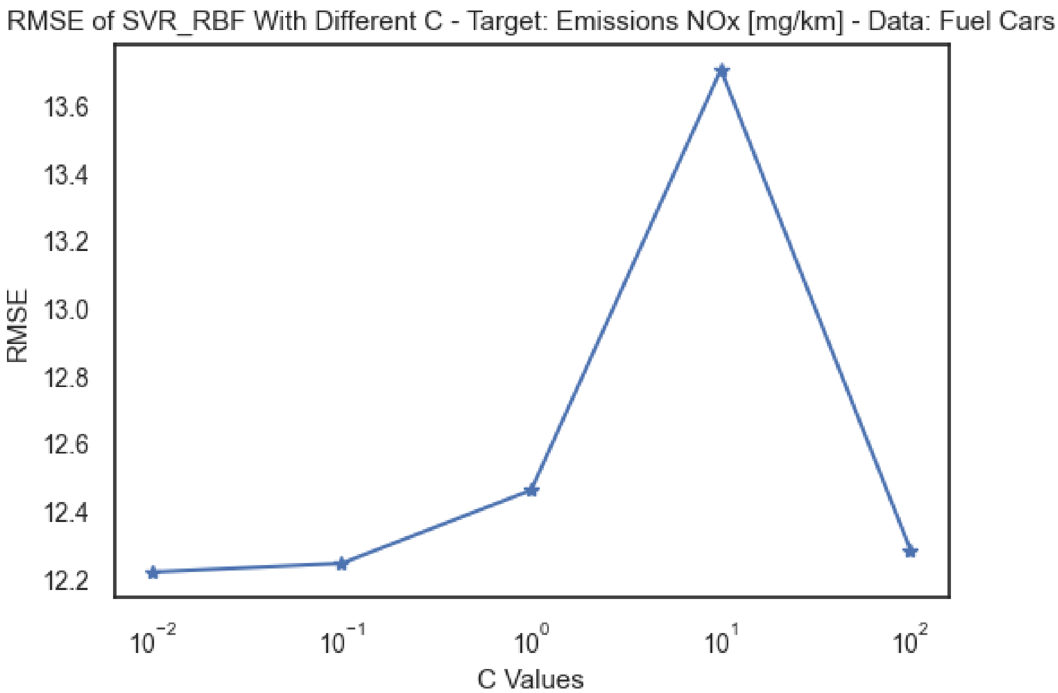
Step2: Using the best C margin, determine the training, test and validation RMSE of the model. This step helps determine how the model adopts to the new data.

Step 3: Aggregate findings in step 2 to determine the best model.

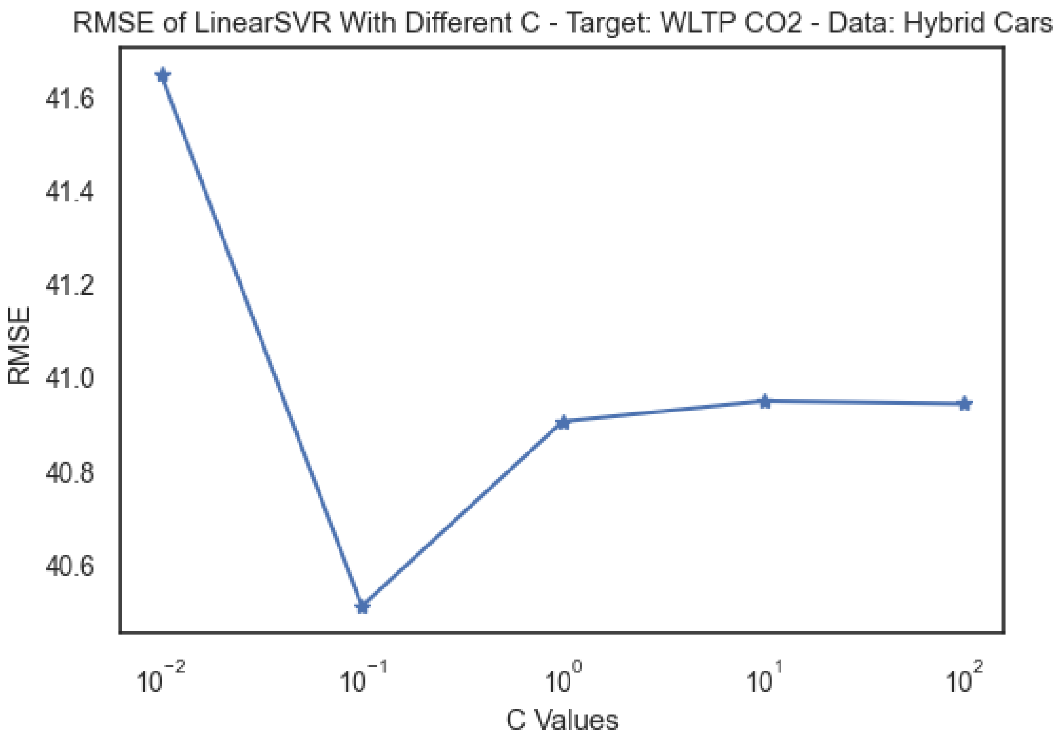


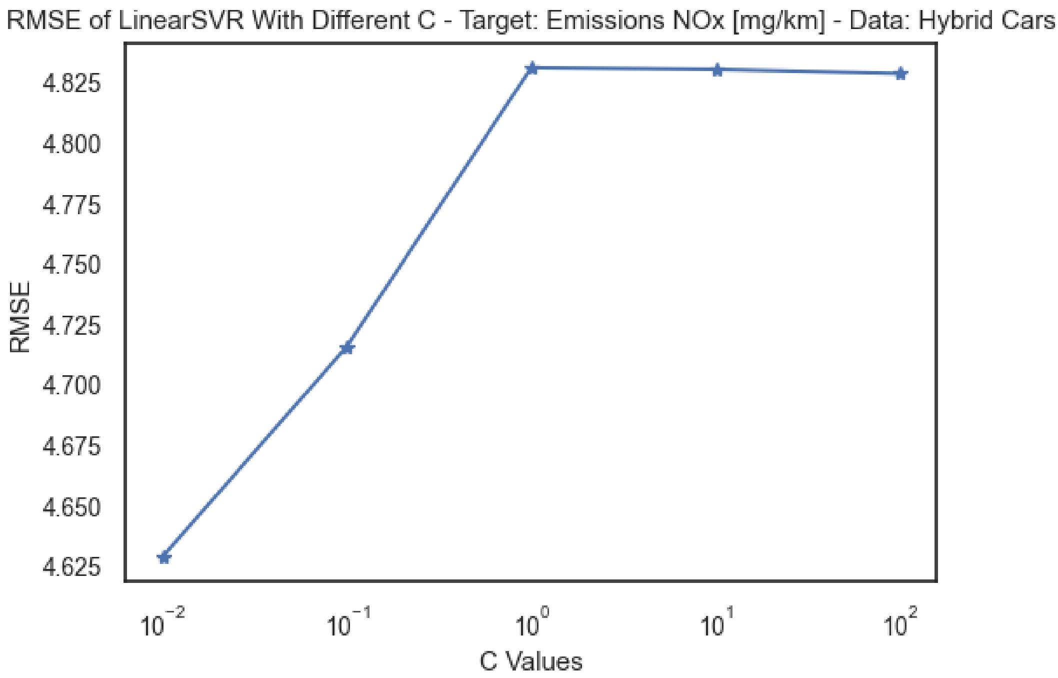

RMSE of LinearSVR With Different C - Target: Emissions NOx [mg/km] - Data: Fuel Cars
RMSE
26
24
22
20
18
16
14
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C Values

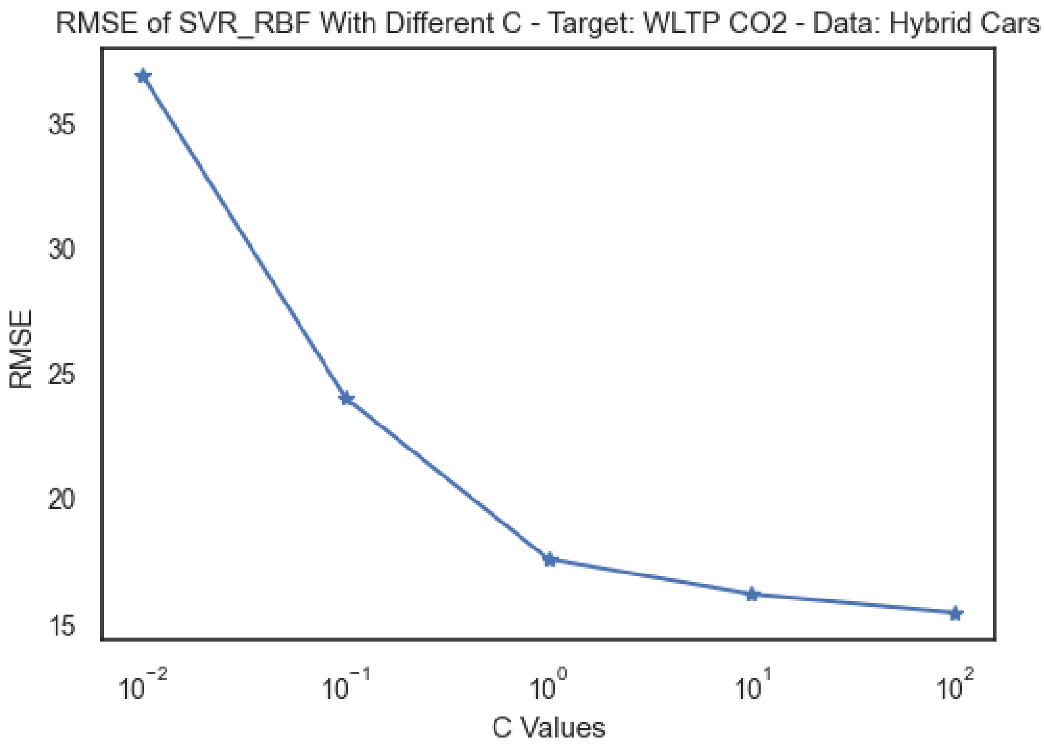

RMSE of SVR_RBF With Different C - Target: WLTP CO2 - Data: Fuel Cars
RMSE
58
56
54
52
50
48
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C Values

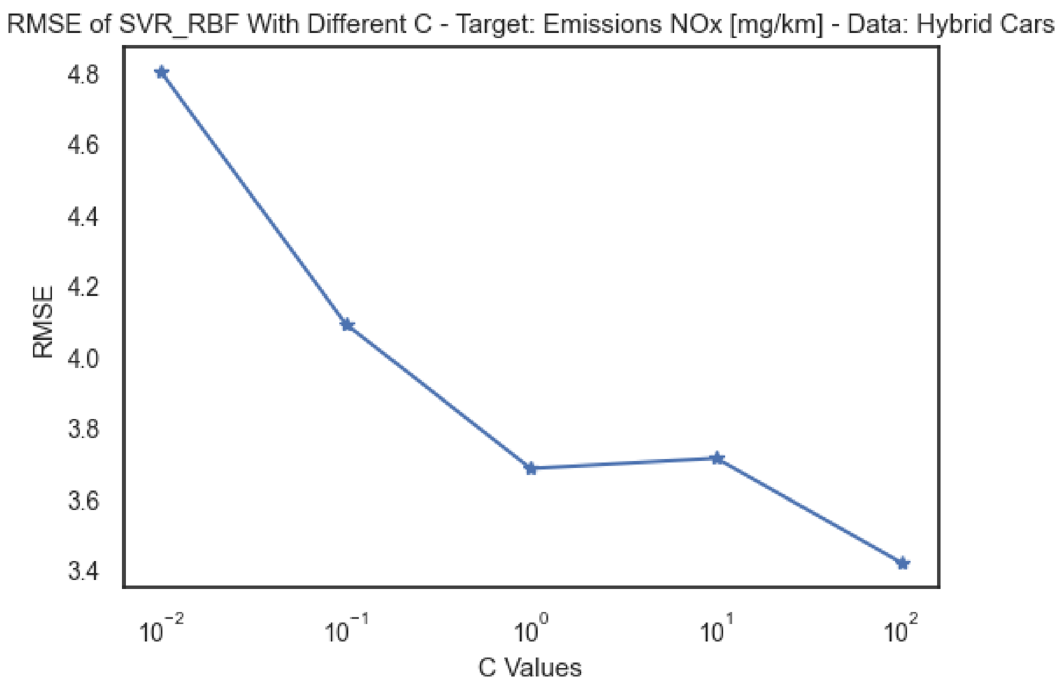


Findings: - Data: Data_Fuel - Target: WLTP CO2 - Model: LinearSVR - Best C: 0.01 - Data: Data_Fuel - Target: Emissions NOx [mg/km] - Model: LinearSVR - Best C: 0.01 - Data: Data_Fuel - Target: WLTP CO2 - Model: SVR_RBF - Best C: 0.01 - Data: Data_Fuel - Target: Emissions NOx [mg/km] - Model: SVR_RBF - Best C: 0.1 - In this setup, RMSE at C - 0.01, 0.1 and 100 are at the same level between 12.2 and 12.3. However, C = 0.1 is selected as it is a balance between the risk of oversimplify (when C = 0.01) and overfitting (when C = 100).


RMSE of LinearSVR With Different C - Target: WLTP CO2 - Data: Hybrid Cars
41.6
41.4
41.2
41.0
40.8
40.6
RMSE
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C Values


RMSE of LinearSVR With Different C - Target: Emissions NOx [mg/km] - Data: Hybrid Cars
4.825
4.800
4.775
4.750
4.725
4.700
4.675
4.650
4.625
RMSE
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
C Values





Findings:
- Data: data_hybrid - Target: WLTP CO2 - Model: LinearSVR - Best C: 0.1
- Data: data_hybrid - Target: Emissions NOx [mg/km] - Model: LinearSVR - Best C: 0.01
- Data: data_hybrid - Target: WLTP CO2 - Model: SVR_RBF - Best C: 100

- Data: data_hybrid - Target: Emissions NOx [mg/km] - Model: SVR_RBF - Best C: 100

[73]:
```
# data_fuel - WLTP CO2 - LinearSVR - C = 0.01
svm_linear_model("WLTP CO2", data_fuel, C= 0.01)
```

```
Training RMSE of the model (linearSVR) is 14.818008428369495
Test RMSE of the model (linearSVR) is 10.671174583695839
Validation RMSE of the model (linearSVR) is 12.106965525400907
```

[74]:
```
# data_fuel - WLTP CO2 - SVM_RBF - C = 0.01
svm_rbf_model("WLTP CO2", data_fuel, C= 0.01)
```

```
Training RMSE of the model (SVR_RBF) is 36.03971978226654
Test RMSE of the model (SVR_RBF) is 48.12935161919517
Validation RMSE of the model (SVR_RBF) is 60.11977282938391
```

Findings 1: Between linearSVR and SVR_RBF models in predicting CO2 emissions from fuel cars, linearSVR is performing better. It records lower RMSEs while the gaps between training RMSE and test RMSE and validation RMSE are also smaller in linearSVR is smaller than the other. Hence, it returns a higher accuracy and a stable performance.

[75]:
```
# data_hybrid - WLTP CO2 - LinearSVR - C = 0.1
svm_linear_model("WLTP CO2", data_hybrid, C = 0.1)
```

```
Training RMSE of the model (linearSVR) is 37.60967196469193
Test RMSE of the model (linearSVR) is 40.51084216883486
Validation RMSE of the model (linearSVR) is 32.31765891375547
```

[76]:
```
# data_hybrid - WLTP CO2 - SVR_RBF - C = 100
svm_rbf_model("WLTP CO2", data_hybrid, C = 100)
```

```
Training RMSE of the model (SVR_RBF) is 16.26323140965809
Test RMSE of the model (SVR_RBF) is 15.438230473931295
Validation RMSE of the model (SVR_RBF) is 11.777385489036144
```

Findings 2: Between linearSVR and SVR_RBF models in predicting CO2 emissions from hybrid cars, SVR_RBF is performing better. The comparisons of figures show that SVR_RBF returns lower RMSEs. Moreover, noting that C in SVR_RBF model is 100. This means this model punishes errors heavily, which, in general, would lead to a higer error. However, the gaps between training, test and validation RMSE of this model are not significant. Hence, this model is outperforming linearSVR in this area.

[77]:
```
# data_fuel - Emissions NOx [mg/km] - LinearSVR - C = 0.01
svm_linear_model("Emissions NOx [mg/km]", data_fuel, C= 0.01)
```

```
Training RMSE of the model (linearSVR) is 15.041403692902918
Test RMSE of the model (linearSVR) is 13.71152606197146
Validation RMSE of the model (linearSVR) is 14.944216879365944
```

[78]:
```
# data_fuel - Emissions NOx [mg/km] - SVM_RBF - C = 0.1
svm_rbf_model("Emissions NOx [mg/km]", data_fuel, C= 0.1)
```

```
Training RMSE of the model (SVR_RBF) is 14.660928513356994
Test RMSE of the model (SVR_RBF) is 12.24657395432253
Validation RMSE of the model (SVR_RBF) is 13.270477753715037
```

Finding 3: Between linearSVR and SVR_RBF models in predicting NOx emissions from fuel cars, SVR_RBF model is a bit better than the other. Given that their RMSE are low and at the same rate (between 12 and 15), C becomes the matter in making decision. A smaller C means a simpler model, hence it has a higher bias. Moreover, a smaller C model more likely allows an error to happen. As a result, SVR_RBF model is selected with C at 0.1.

[79]:
```
# data_hybrid - Emissions NOx [mg/km] - LinearSVR - C = 0.1
svm_linear_model("Emissions NOx [mg/km]", data_hybrid, C = 0.1)
```

```
Training RMSE of the model (linearSVR) is 6.894769288588417
Test RMSE of the model (linearSVR) is 4.715904890092227
Validation RMSE of the model (linearSVR) is 8.769208267487658
```

[80]:
```
# data_hybrid - Emissions NOx [mg/km] - SVR_RBF - C = 100
svm_rbf_model("Emissions NOx [mg/km]", data_hybrid, C = 100)
```

```
Training RMSE of the model (SVR_RBF) is 4.9344343500396155
Test RMSE of the model (SVR_RBF) is 3.4225757264201553
Validation RMSE of the model (SVR_RBF) is 8.097441934502436
```

Finding 4: Between linearSVR and SVR_RBF models in predicting NOx emissions from hybrid cars, linearSVR model outperforms the other. The outcomes record that both have relevantly small RMSE accrosing training, test and validation dataset. however, the notice is at the gap between test RMSE and validation RMSE in SVR_RBF model, which is significant (3.4 and 8.90). This suggests an inconsistent of performance of this model with new data. As a result, linearSVR is chosen in this comparison.

Conclusion in training SVM models:

In the model selection process, two main criteria are considered:

- Model accuracy using RMSE metric.
- Model complexity based on C value

As a result, there are four models selected for different set of target variable and dataset.
1. LinearSVR with C at 0.01 is the best in predicting CO2 emission from fuel cars (Finding 1)
2. SVR_RBF with C at 100 is the best in predicting CO2 emission from hybrid cars (Finding 2)
3. SVR_RBF with C at 0.1 is the best in predicting NOx emission from fuel cars (Finding 3)
4. LinearSVR with C at 0.1 is the best in predicting NOx emisison from hybrid cars (Finding 4)

6 Conclusion

For CO2 Emissions prediction from fuel cars - Within the same target variable and dataset, in comparison to Neural Network Regression (4 Layers) above, LinearSVR with C at 0.01 is the best in predicting CO2 emission from fuel cars due to having lower gaps of RMSE. The RMSE of training set is also lower than the figure for test set.

For CO2 Emissions prediction from hybrid cars - In comparison between Neural Network (3 layers) and SVR_RBF model, there are no significant differences in RMSE values. Hence, both can be kept.

For NOx Emissions prediction from fuel cars and hybrid cars - As there is no Neural Network Regression for comparison, two models in Finding 3 and 4 of SVM model training are kept as the only best models. - They are - SVR_RBF with C at 0.1 is the best in predicting NOx emission from fuel cars (Finding 3) - LinearSVR with C at 0.1 is the best in predicting NOx emisison from hybrid cars (Finding 4)

7 References

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